Bell Time & Walk Zone Analysis Report

Prepared for

Brecksville – Broadview Heights City School District
Brecksville, OH

Prepared December 10, 2021

Submitted by:

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Project Summary

Transfinder Professional Services was contracted to provide analysis on Brecksville - Broadview Heights City School District's (BBHCSD) current bus routes, as well as to explore proposed changes to school Bell Time and Walk Zones. The Bell Time option to be analyzed involves switching from the current three-tiered model to a two-tiered model. In preparation for the proposed change, BBHCSD transportation staff has prepared routes that combine the Middle School and High School routes into a single tier and the Elementary routes to deliver students to a new elementary school currently being built. Using current routes as a baseline for comparison with the modified routes, this report reviews the proposed option for BBHCSD to change their school bell times.

Current Bell Times:

Brecksville - Broadview Heights City School District currently has one High School, one Middle School and three Elementary Schools. The following chart shows current bell times with their tiers of service in the morning and afternoon.

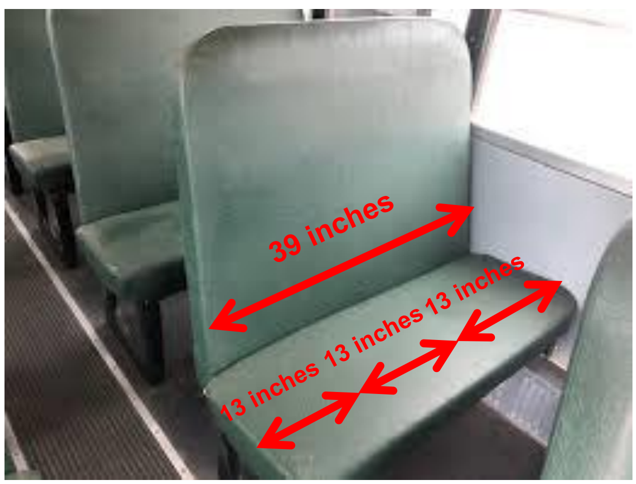
School Names	Arrive	AM Bell	PM Bell	Depart	Length of Day
High School	7:15	7:35	2:35	2:42	420 min
Middle School	8:00	8:10	3:05	3:15	415 min
Chippewa Elem School	8:55	9:10	3:40	3:50	400 min
Highland Elem School	8:55	9:10	3:40	3:50	400 min
Hilton Elem School	8:55	9:10	3:40	3:50	400 min

Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) are nationally recognized measurements of your current transportation operation. KPIs allow you to compare your operational performance year after year. They can also provide a comparison of your own performance in relation to other school districts of similar size, if that information is available. This report provides the baseline for you to use in further comparisons.

Ridership KPI

The typical school bus seat is 39 inches wide. This allows for roughly 13 inches of seating space for each student. This works well for smaller elementary students, however, typically only two secondary education students will be able to sit on a school bus seat.



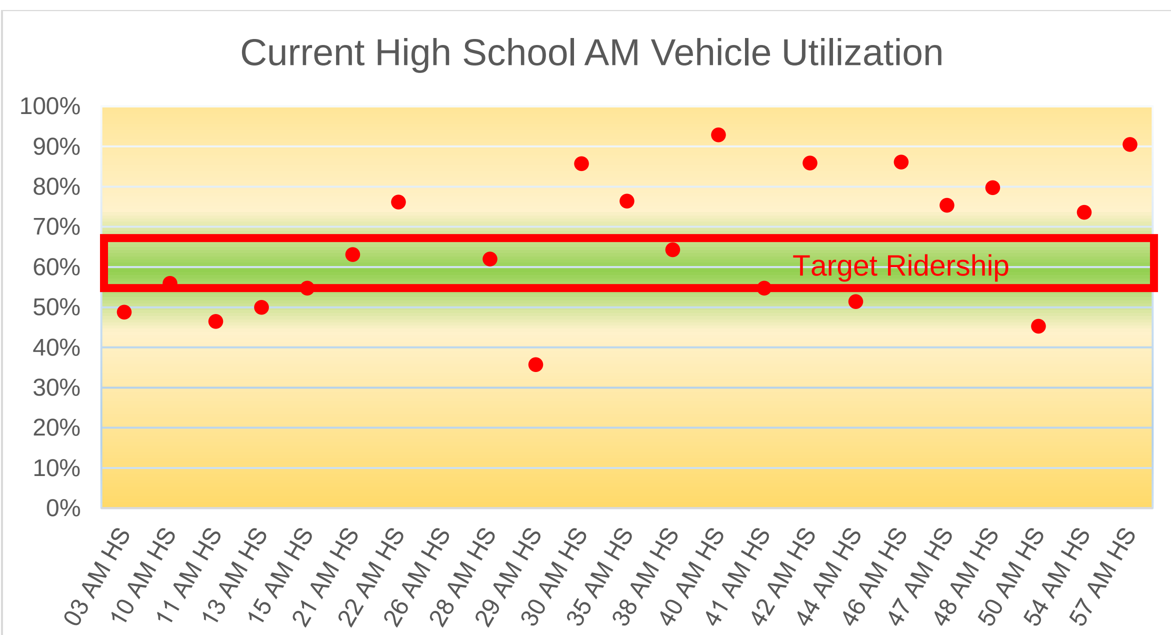
The manufacturer's rated capacity places three students on every bus seat which is not possible for most middle school and high school students.

Ability to place additional students on the buses, and student ride time form the basis for routing efficiency. The charts on the following pages show current utilization of individual buses by tier based on manufacturer's rated capacity for each bus and the number of students riding the buses. Buses are shown on the horizontal axis and percent of rated capacity is shown on the vertical axis. An acceptable efficiency range for secondary is between 55% and 70% for actual riders. Likewise, the efficiency range for elementary runs is between 60% and 80% of capacity for actual riders. Sixty-six percent of capacity is 2 riders per seat and 80% is approximately 3 riders per seat for ½ the bus and 2 riders per seat for the rest.

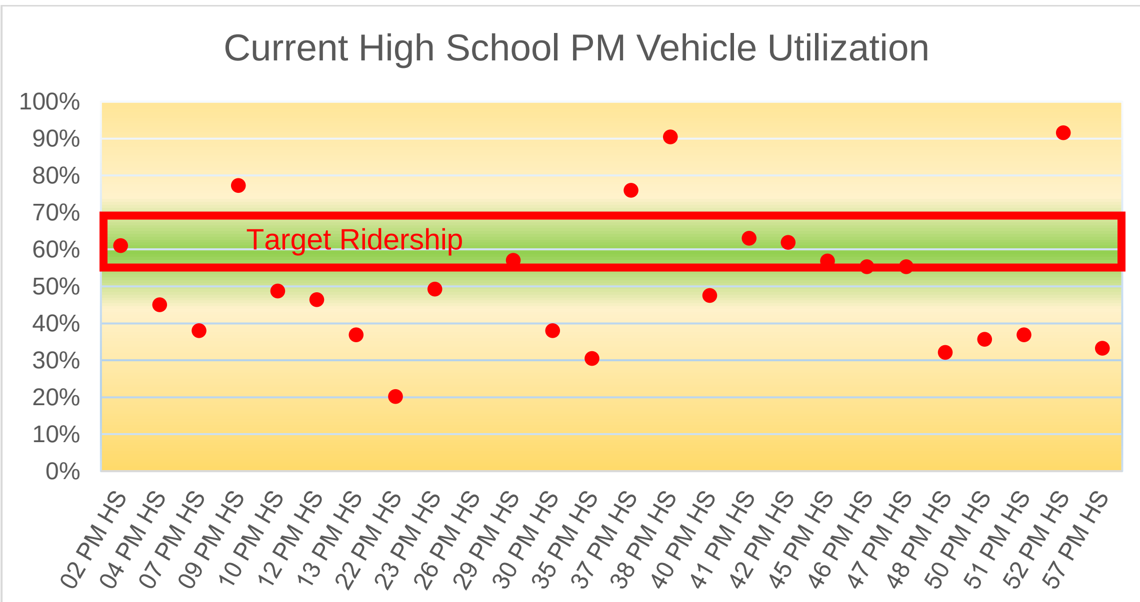
These efficiency ridership targets are taken from guidance provided by the Council on Great City Schools as well as the National Association of Pupil Transportation. Additional information on efficiency targets can be found in the book, “Best Practices in Student Transportation” written by Dan Roberts.

Due to the Covid-19 pandemic, BBHCSD has set targets for ridership on the elementary buses to allow for a little more space between students. Their target has been set between 55% and 70% to keep the number of students per seat around 2.

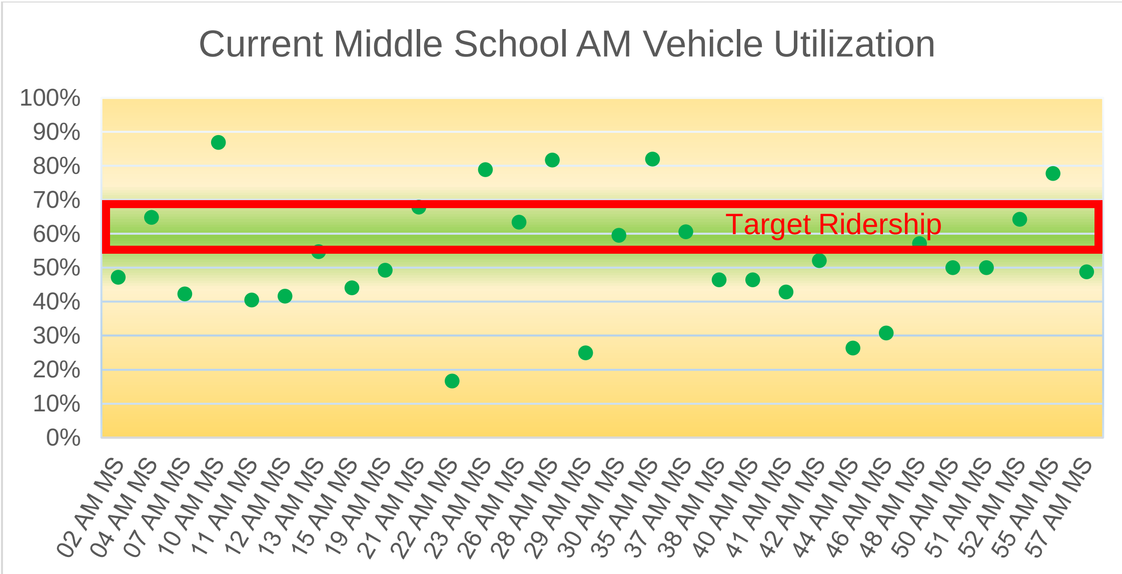
To form a baseline for comparison, the number of assigned riders from district 2021 -2022 records was compared to the capacity of the buses. This can provide insight into whether or not there may be efficiencies that can be gained with increased vehicle utilization.



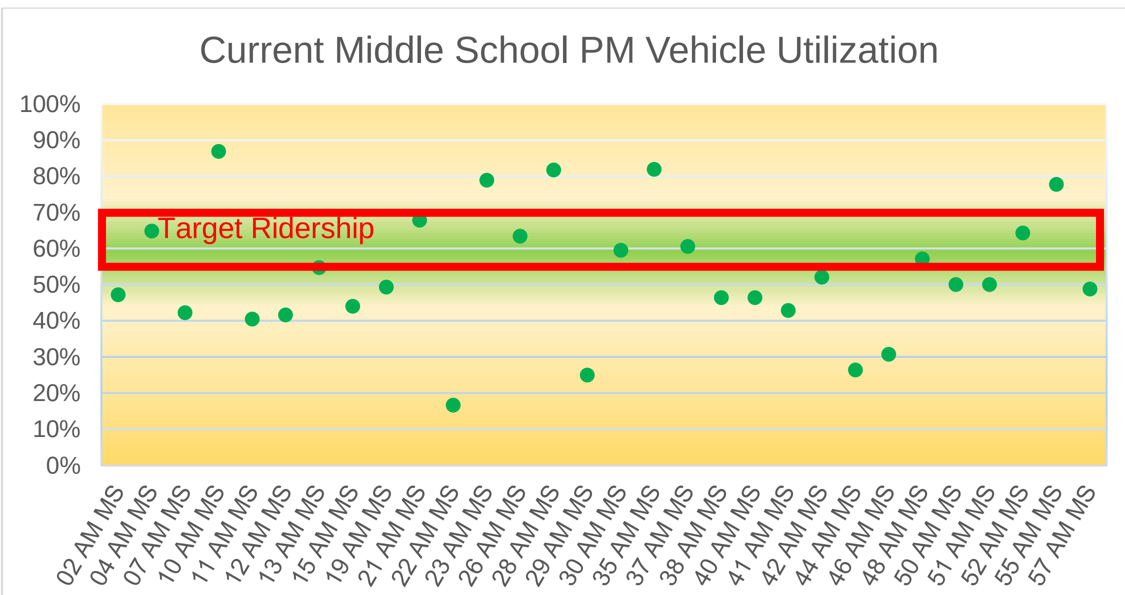
Ridership at the high school level often fluctuates on a day to day basis. This is especially true in the afternoon when students have different after school programs in which they may participate. As a result, many transportation departments leave a little extra space on the buses to handle this fluctuation. Typically this results in ridership levels that are well below the target rideship of the vehicle capacity. Average ridership across all BBHCSD High School bus runs is 61%. This is pretty high given the small amount of time between tiers, especially for the afternoon dismissal.



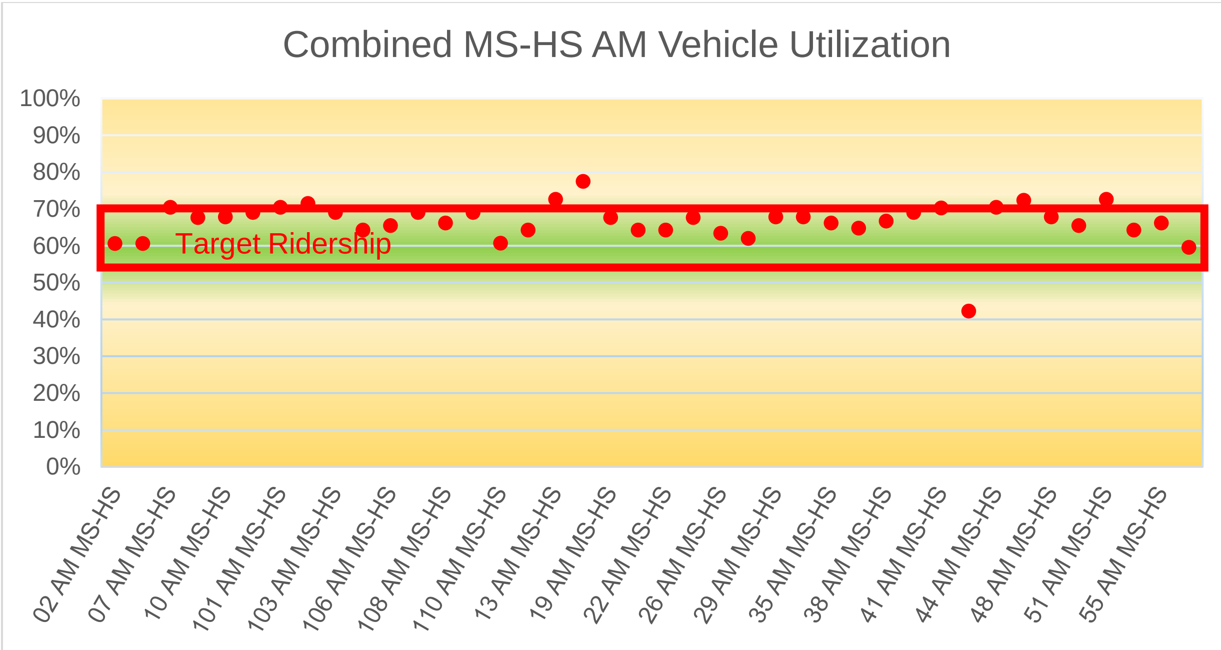
Below are similar charts for the current Middle School routes showing ridership compared to vehicle utilization. Though there is slightly less fluctuation of ridership at the middle school level there is also an impact from the amount of time available between the bells of the other schools. This lack of time often leads to slightly more under-utilized vehicles because there simply isn’t time to transport a full bus.



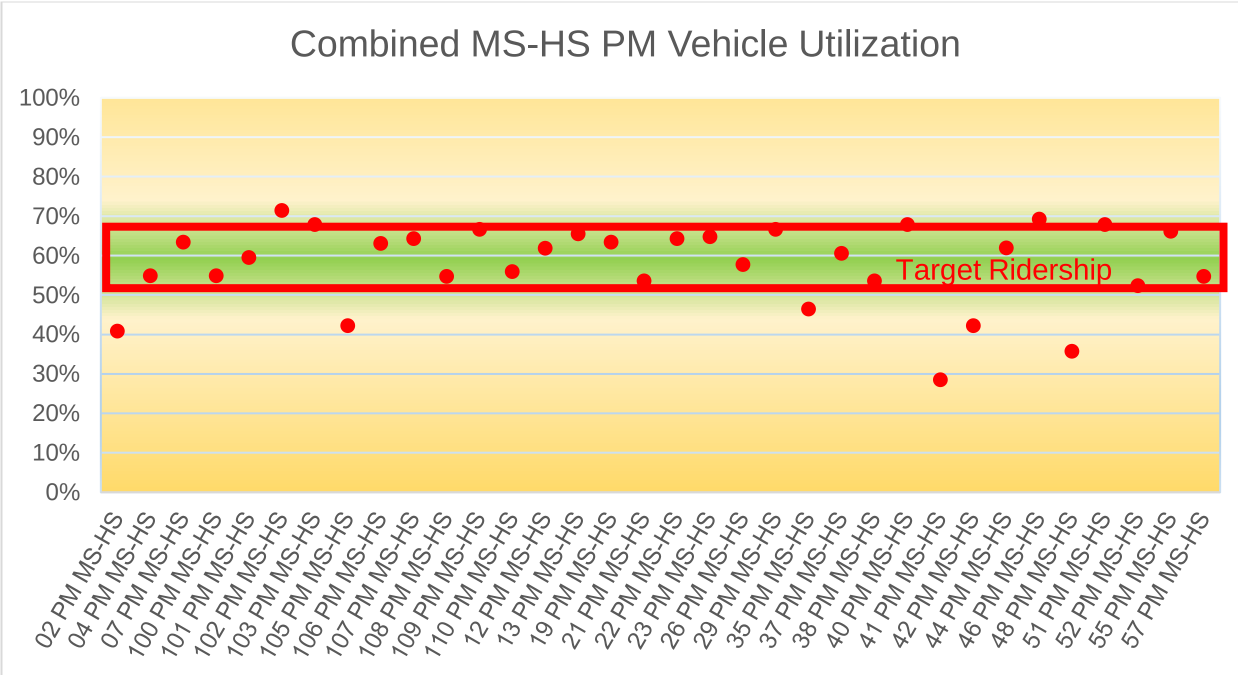
Average ridership across all Middle School bus runs is 53%. This is pretty high given the small amount of time between tiers, especially for the afternoon dismissal.



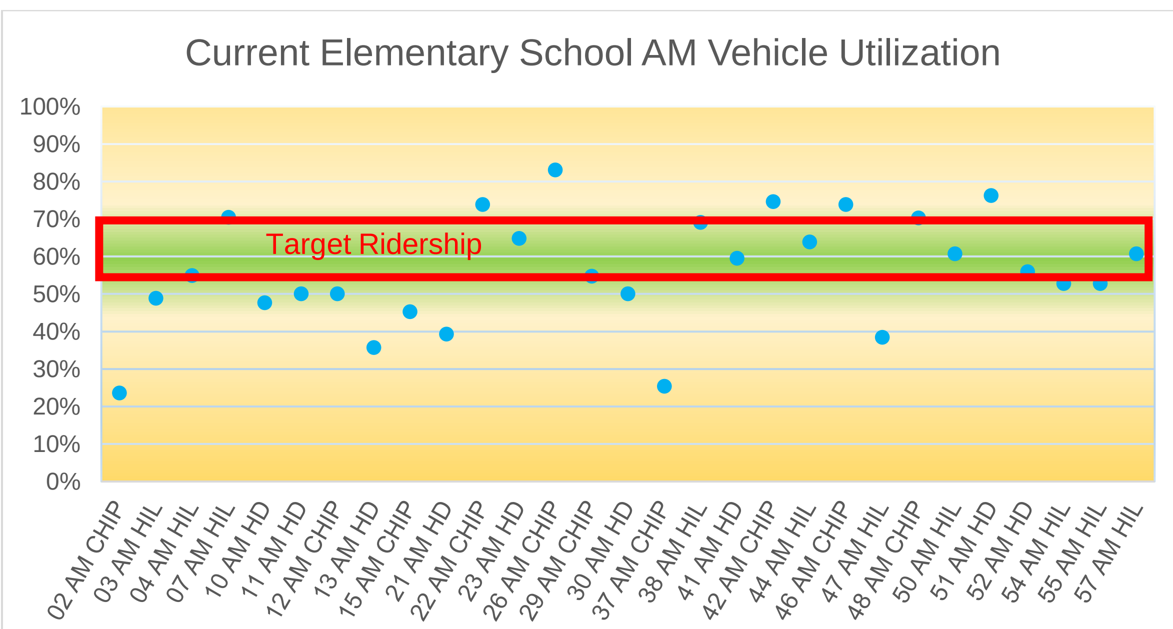
For comparison, below are utilization charts for the planned combined middle school and high school routes.



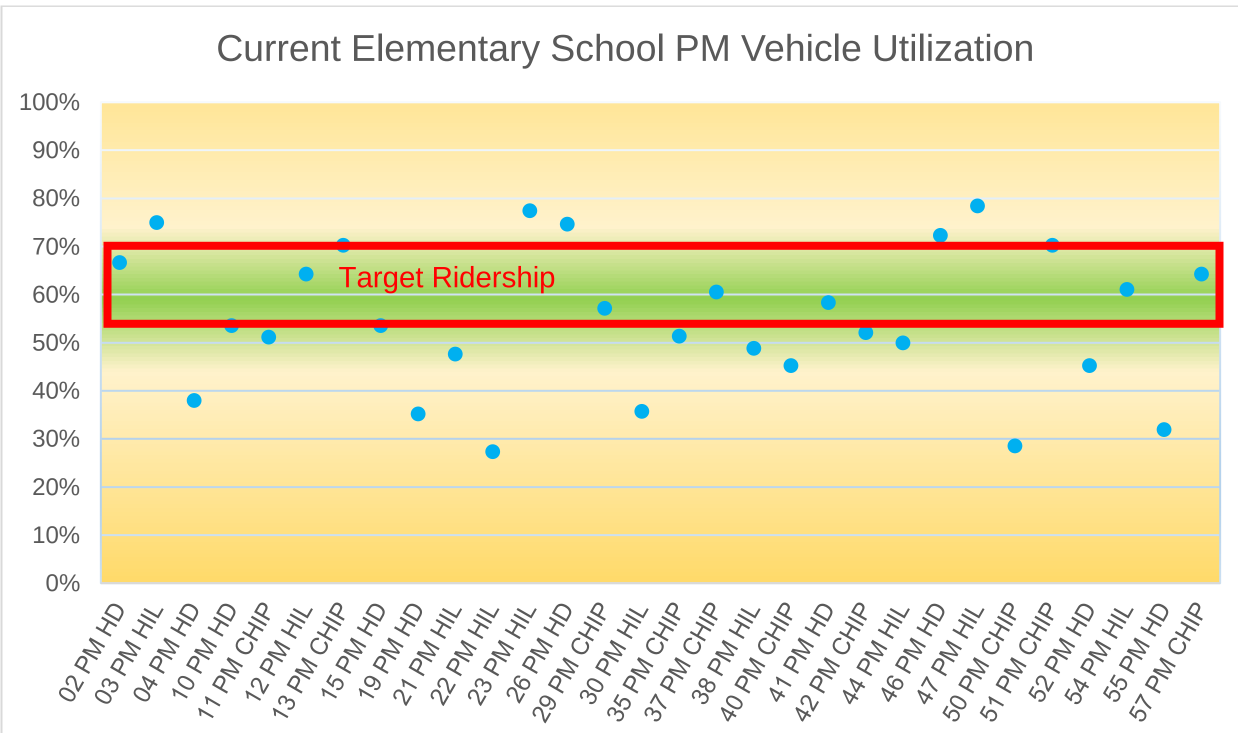
These routes are based on projections of the students likely to reside in the areas covered by the planned routes. As students move in and out of the community, or as they move within the community, the ridership on buses will fluctuate. Each year transportation adjusts routes to the fluctuating enrollment and as such these planned routes will change before they are implemented next year.

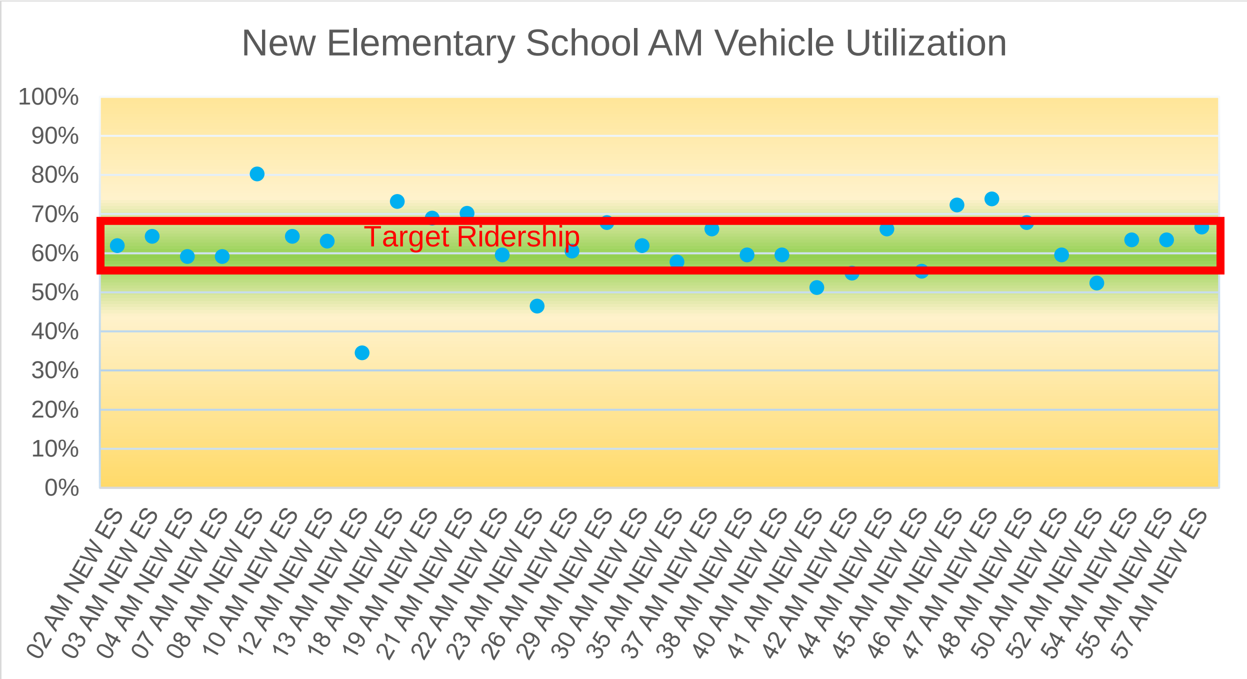


Similar to the ridership charts for the Middle School and High School above, you will find ridership charts for Elementary on the following pages. For comparison, you will find charts for the current Elementary Schools and for the planned routes for the new elementary school.



Average ridership across all Elementary School bus runs is 56%. Though it might be possible to have higher utlization on the current Elementary buses, concerns regarding spacing for students relative to the current Covid-19 pandemic have warranted lower utilization. Additionally, it is the buses with lower ridership and shorter run times that can be easily re-utilized when drivers call in sick and there aren't enough substitures to cover all bus runs. Eliminating this flexibility would impact transportation's ability to dispatch buses to handle extra bus runs or emergency situation.




New Elementary School AM Vehicle Utilization
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Target Ridership
02 AM NEW ES
03 AM NEW ES
04 AM NEW ES
07 AM NEW ES
08 AM NEW ES
10 AM NEW ES
12 AM NEW ES
13 AM NEW ES
18 AM NEW ES
19 AM NEW ES
21 AM NEW ES
22 AM NEW ES
23 AM NEW ES
26 AM NEW ES
29 AM NEW ES
30 AM NEW ES
35 AM NEW ES
37 AM NEW ES
38 AM NEW ES
40 AM NEW ES
41 AM NEW ES
42 AM NEW ES
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47 AM NEW ES
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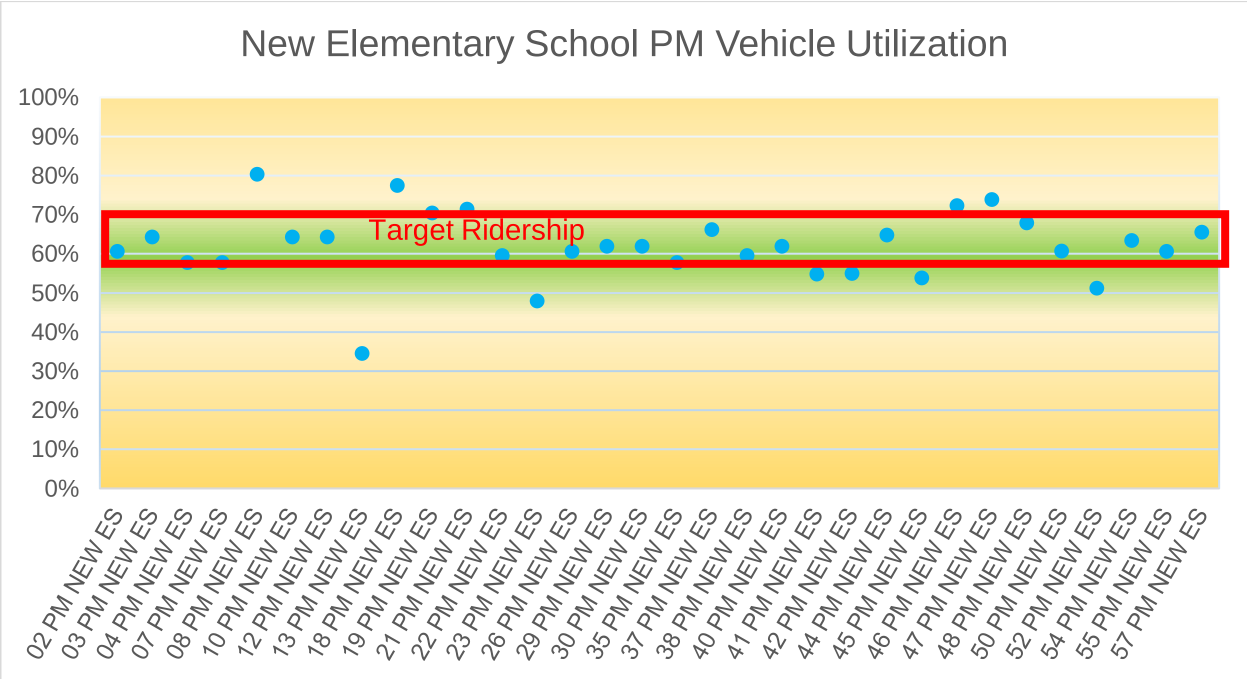
55 AM NEW ES
57 AM NEW ES


New Elementary School PM Vehicle Utilization
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Target Ridership
02 PM NEW ES
03 PM NEW ES
04 PM NEW ES
07 PM NEW ES
08 PM NEW ES
10 PM NEW ES
12 PM NEW ES
13 PM NEW ES
18 PM NEW ES
19 PM NEW ES
21 PM NEW ES
22 PM NEW ES
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57 PM NEW ES

Reviewing the charts above for the current three-tiered schedule, it should be noted that in each tier there are a few buses that are relatively underutilized. For example, Bus 22 has low ridership in each tier of the afternoon routes. Given the driver shortage that nearly all school districts are facing, in addition to the many days when multiple drivers call in sick, there may not be enough drivers to cover all routes. These under-utilized routes are usually the ones that can be dispatched to cover routes without a driver on those days. Without this flexibility, those routes missing a driver would be considerably late.

The table below shows BBHCSD enrollment on December 8, 2021.

ENROLLMENT COUNT SY 21/22

Run Date 12/8/21	**KG**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Total**
Chippewa	86	82	109	82	87	446
Highland	86	87	69	75	73	390
Hilton	79	91	77	87	82	416
Middle School	**Grade 5**	**Grade 6**	**Grade 7**	**Grade 8**		
	243	253	280	307		1083
High School	**Grade 9**	**Grade 10**	**Grade 11**	**Grade 12**	**Grade 23**	
	302	304	297	297	4	1204
					District Total	**3539**

The table on the following page shows the number of current trips that could be reduced with unlimited time. **Time is a critical factor, in order to fully utilize vehicles, there must be enough time to pick up a full load of students.**

Schools	Total Students	Assigned Riders	Current Runs	Target Riders/Bus	Potential Runs	Delta
High School	1204	1103	26	48	23	3
Middle School	1083	1194	31	48	25	6
Elementary Schools	1252	1186	30	48	25	5

Note: In the three-tiered schedule many private school students ride during the Middle School tier, making the total number of students assigned during that tier higher than the number of enrolled students.

For comparison, the table below shows the same information for the proposed two-tiered schedule.

Schools	Total Students	Assigned Riders	Current Runs	Target Riders/Bus	Potential Runs	Delta
Middle School/High School (& private schools)	2161	2084	40	48	43	-3
Elementary Schools	1624	1515	32	48	32	0

The negative three delta for the number of runs in the chart above is an indication that some buses are routed just lightly over the target capacity of 48 students per bus. This is efficient routing for the planned set of Middle School and High School routes using the conservative approach of routing all possible students.

It should also be noted that the grade configuration of the Middle School and the new elementary school will change. The fifth grade will now be in the new elementary school. This increases the elementary population and decreases the middle school population. The planned set of routes for the new elementary school adjust the current routes and only add two buses for the additional fifth grade students. Again, this is efficient routing for the planned set of new elementary school routes using the conservative approach of routing all possible students.

Similar to the current three-tiered schedule, the plan is for private school students to ride during the Middle School and High School tier. This causes the number of assigned riders to be higher than the total enrollment of those two schools.

In the three-tiered schedule the largest population of students to be transported was at the Middle School, thus requiring the largest number of vehicles. Even though the fifth grade is no longer part of the Middle School population, with the Middle School and High School tiers combined, this number increases considerably.

It remains to be seen how the changes in school configurations, and the ongoing pandemic, will impact student ridership on the bus. At this stage it is best to plan routes assuming that practically all students will ride the bus. The proposed set of routes does this.

The planned set of routes for the 2022-2023 school year are all planned for two students per seat, including at the new elementary school. This is being done to ensure extra space due to Covid-19 concerns. Should the pandemic subside, it may be possible to shift the elementary routes to have a higher vehicle utilization closer to the recommended 80% vehicle utilization mention on page 4 which is approximately 3 riders per seat for ½ the bus and 2 riders per seat for the rest.

It is recommended that BBHCSD track ridership on the bus routes after they are implemented in the 2022-2023 school year in order to determine the likely number of riders for each grade level. Though each district has differing levels of ridership in their community, the national average is that roughly 80% of potential elementary students that may ride actually do ride the bus. Similarly, an average of 70% of middle school students who may ride the bus actually ride the bus, and between 50-60% of high school students who may ride the bus actually ride the bus. Taking attendance on the buses next year would provide BBHCSD with more data that may allow further reductions in the number of buses needed.

The table on the following page highlights how the number of actual students riding can vary between grade levels. The total and assigned number of private school students has also been added for reference.

Schools	Total Students	Assigned Riders	Ave % of students riding	Likely Riders
High School	984	929	60%	558
Middle School	934	910	70%	446
Elementary Schools	1624	1515	80%	1212
Private Schools	243	239		

Below is a modified version of the table shown on the bottom of page 11 that shows the potential number of vehicles that might be needed if you were to apply this concept of the likely number of riders to the planned set of routes. This also reflects a higher vehicle utilization for the elementary bus runs which would be 80% utilization.

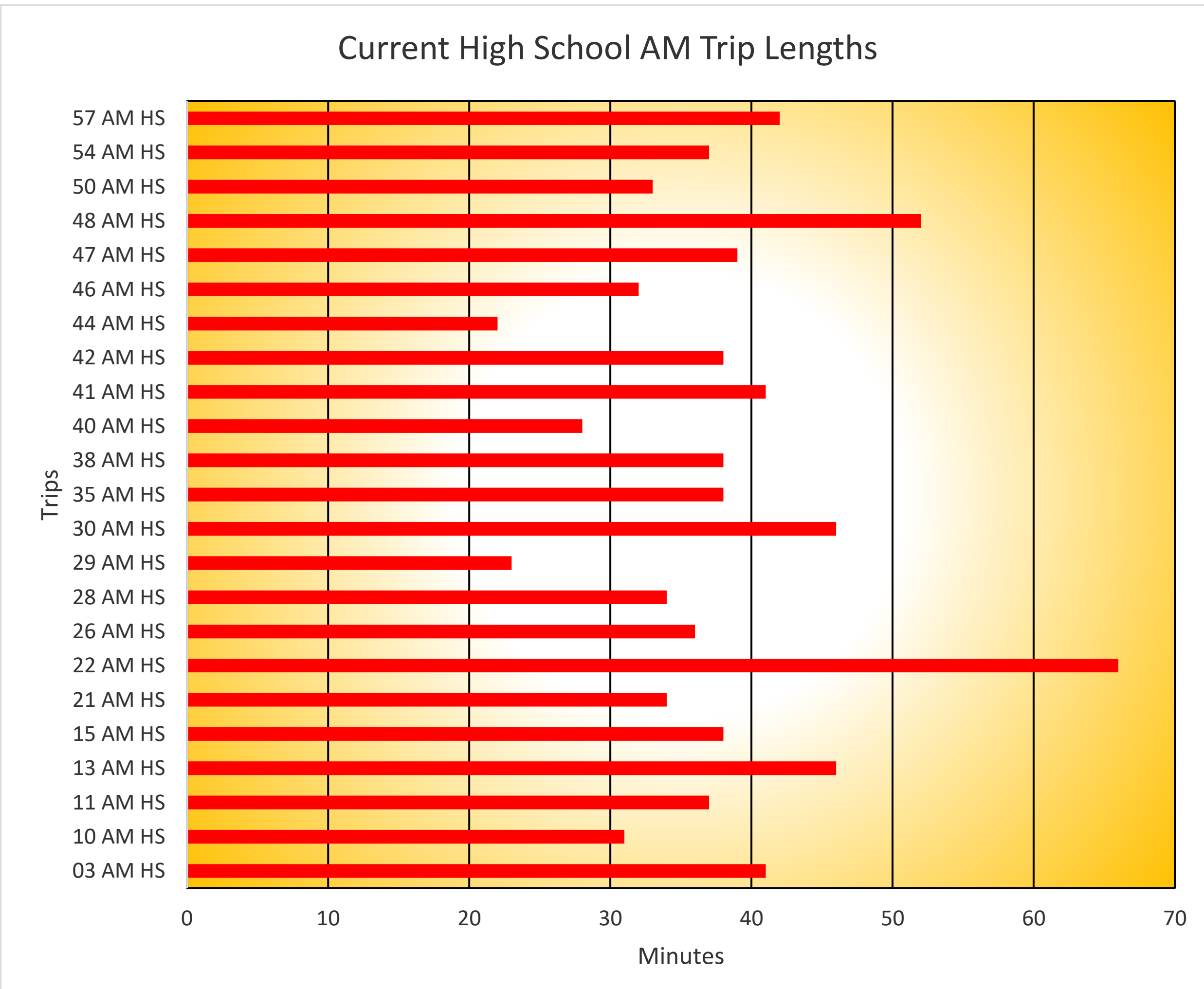
Schools	Total Students	Likely Riders	Planned 22-23 Runs	Target Riders/Bus	Potential Runs	Delta
Middle School/High School (& private schools)	2161	1243	40	48	26	14
Elementary Schools	1624	1212	32	56	22	10

After tracking ridership with the new two-tiered routes for 2022-2023 it is found that ridership at BBHCSD is similar or lower than the national averages it appears likely that reductions in the number of vehicles needed to transport students would be possible. Given unlimited time to fill the vehicles to target ridership it may be possible to route regular education students with as few as 26 buses. As you will see later in this document, this does not yet include other vehicle needs like special needs routes, preschool routes, etc.

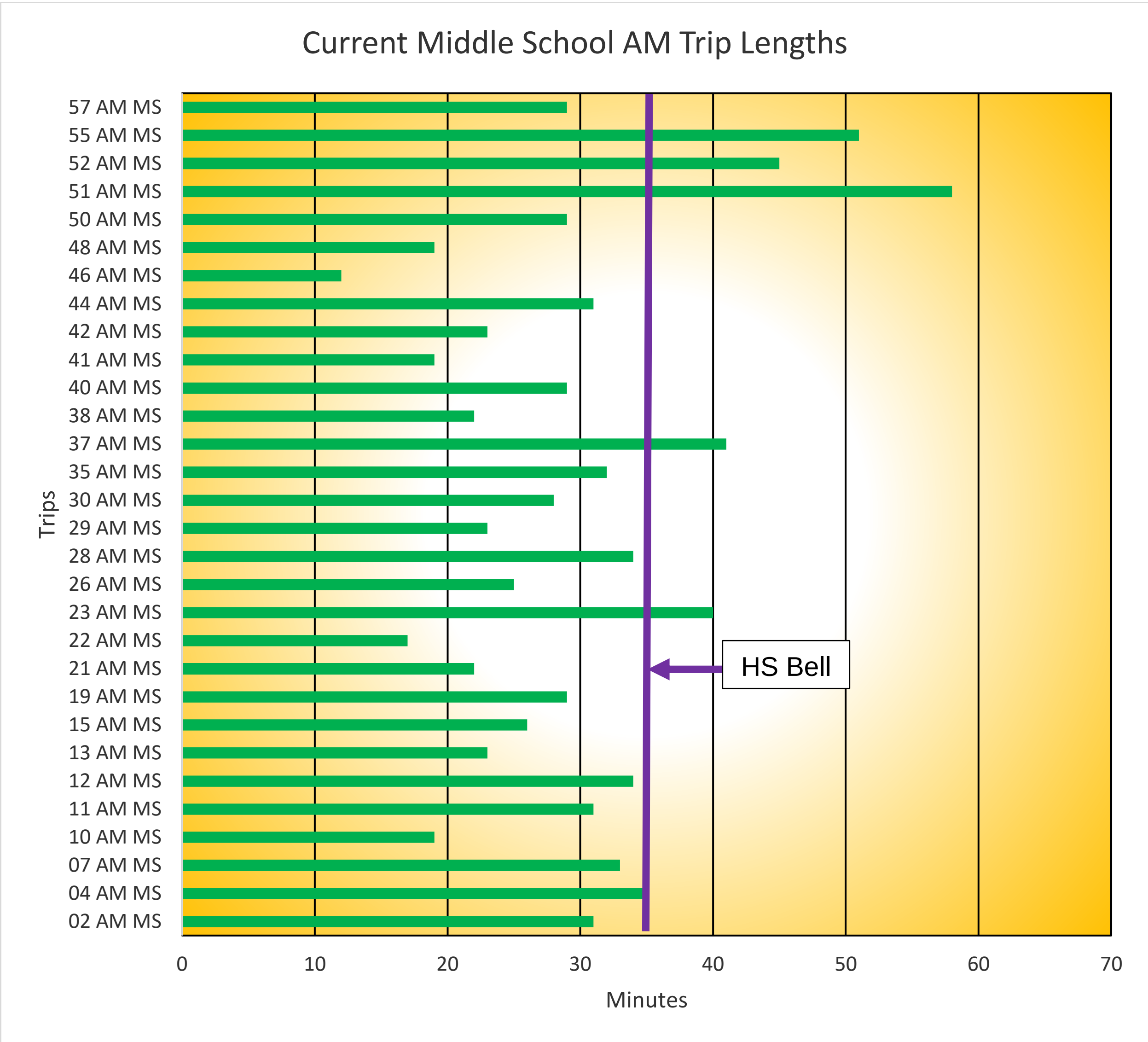
Length of Trip KPI

The length of the individual trip compared to the spacing between bell times (tiers) is a measurement of on-time service and/or the ability to efficiently utilize buses. Buses should have time to complete their trips and arrive at the next tier of service before the next arrival or dismissal bells.

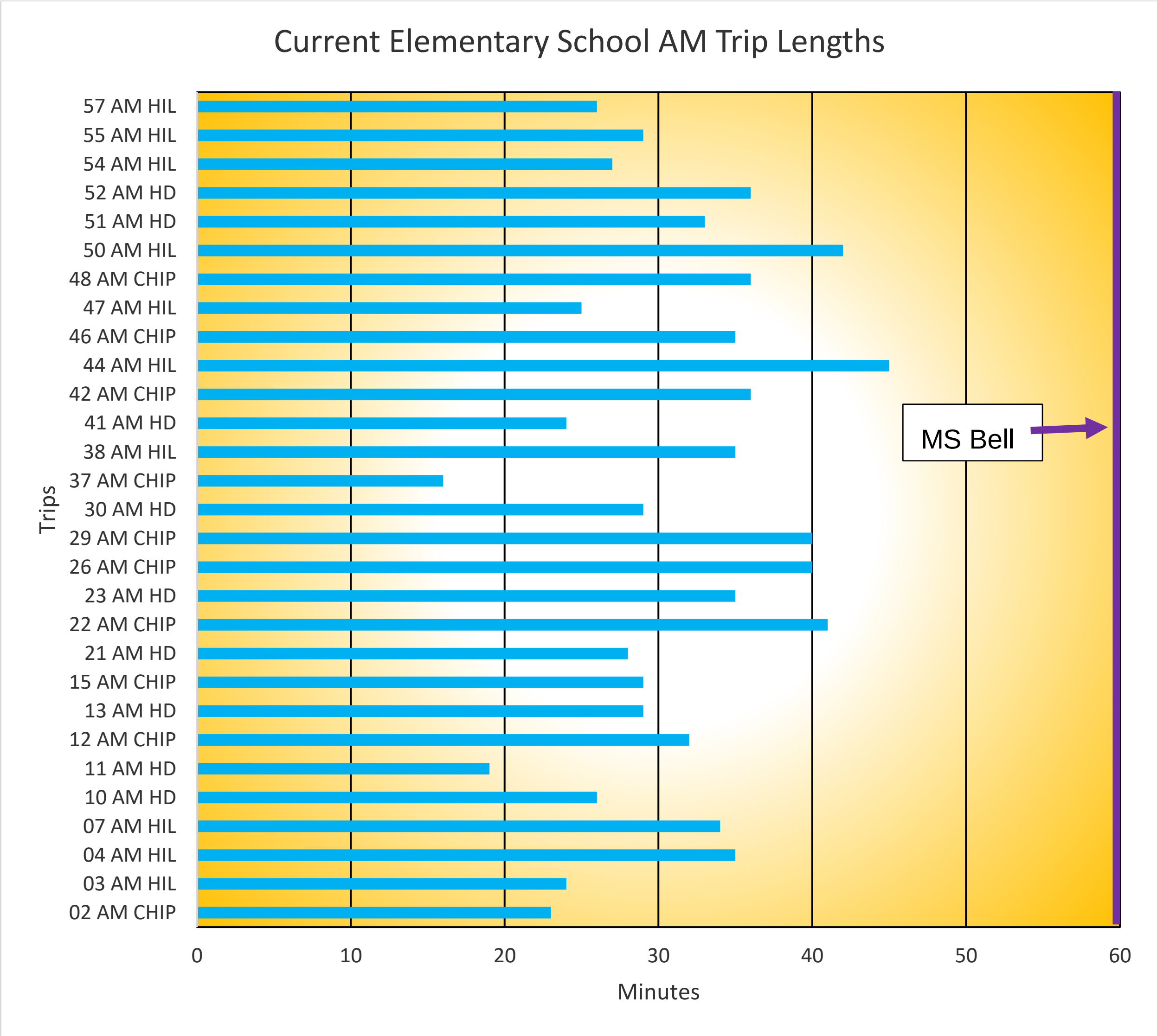
The following charts show the trip times utilizing the current route structure. In most instances there are 35 minutes or more between the morning bell times, **with the ability to drop students off early**. This provides ample time to allow buses to complete their runs with few time conflicts. Time conflicts are illustrated by a purple vertical line on any of the charts below.



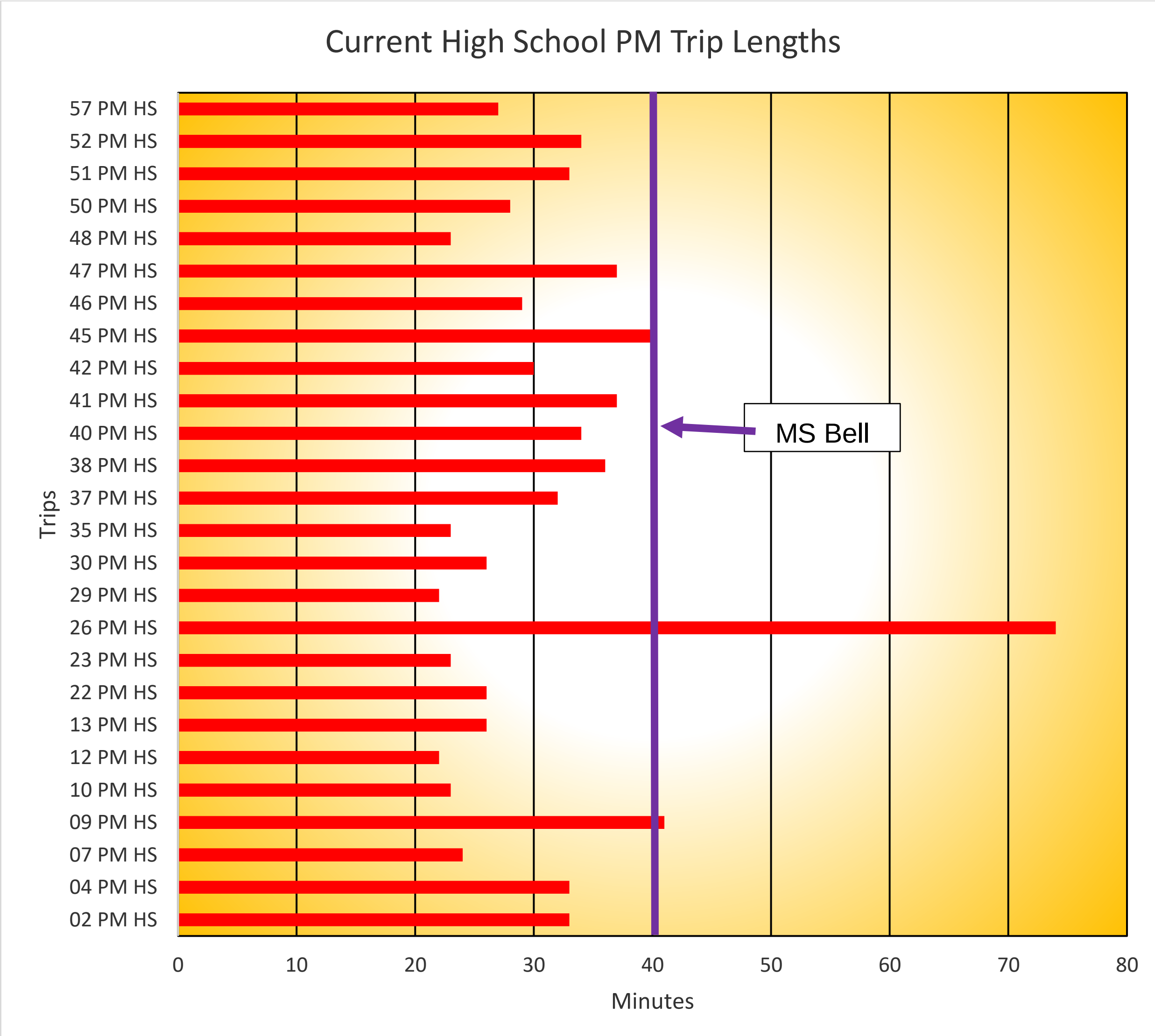
In the morning, if the first trip a bus takes is longer, the bus can leave earlier to deal with the longer run time. However, subsequent trips with longer times can cause conflicts with the bell times. Longer trips will need to start prior to the previous tier's bell time. This typically requires additional vehicles. **The purple line in the chart on the following page illustrates the bell time of the previous tier.**



There are no Elementary runs in the morning that require more time than the amount between bells.

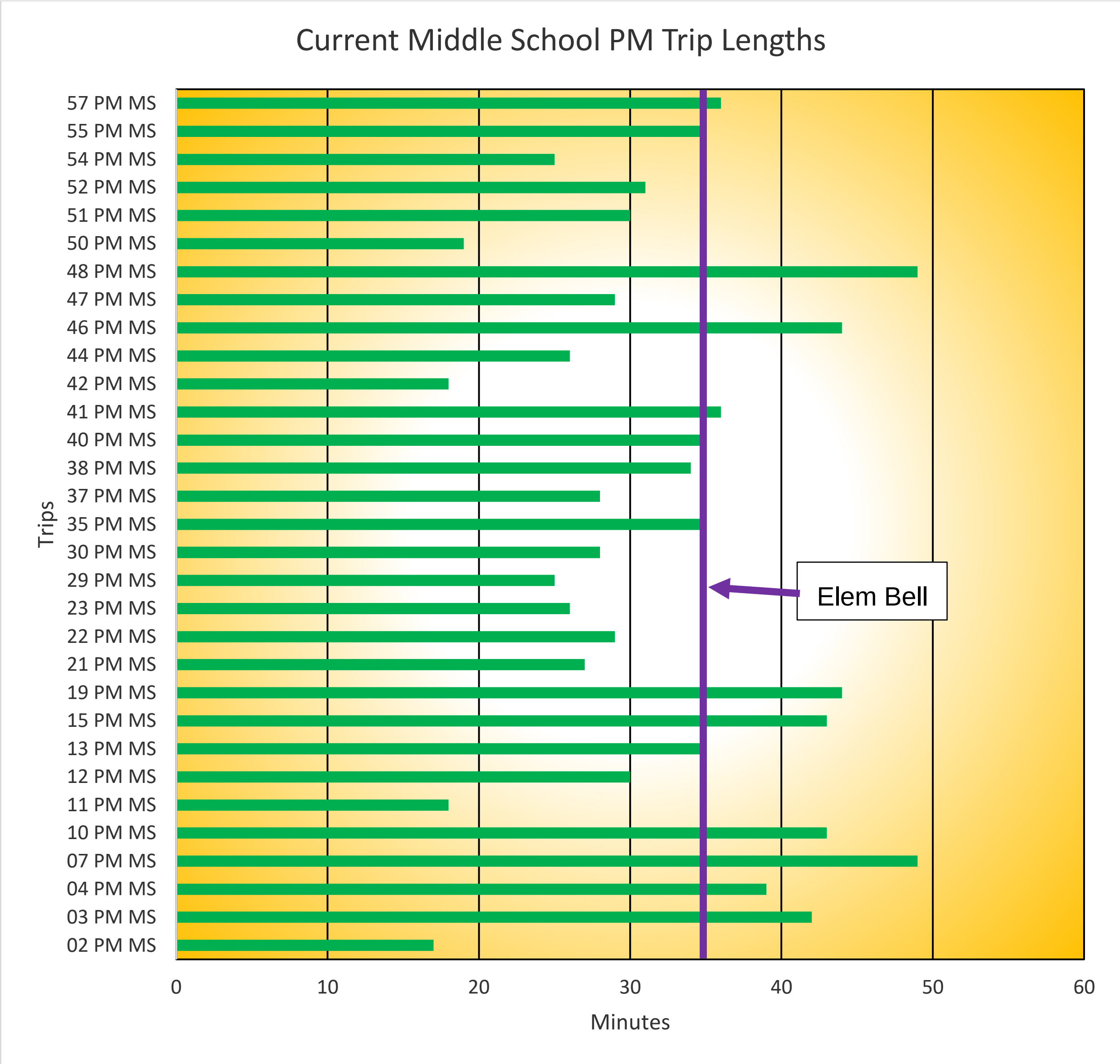


In the afternoon, the first tier of service must start at a specific time at the school being served. This means that longer trips at this tier can cause conflicts with subsequent tiers. As was shown with the AM runs, the purple lines in the charts indicated potential conflicts of service between tiers.



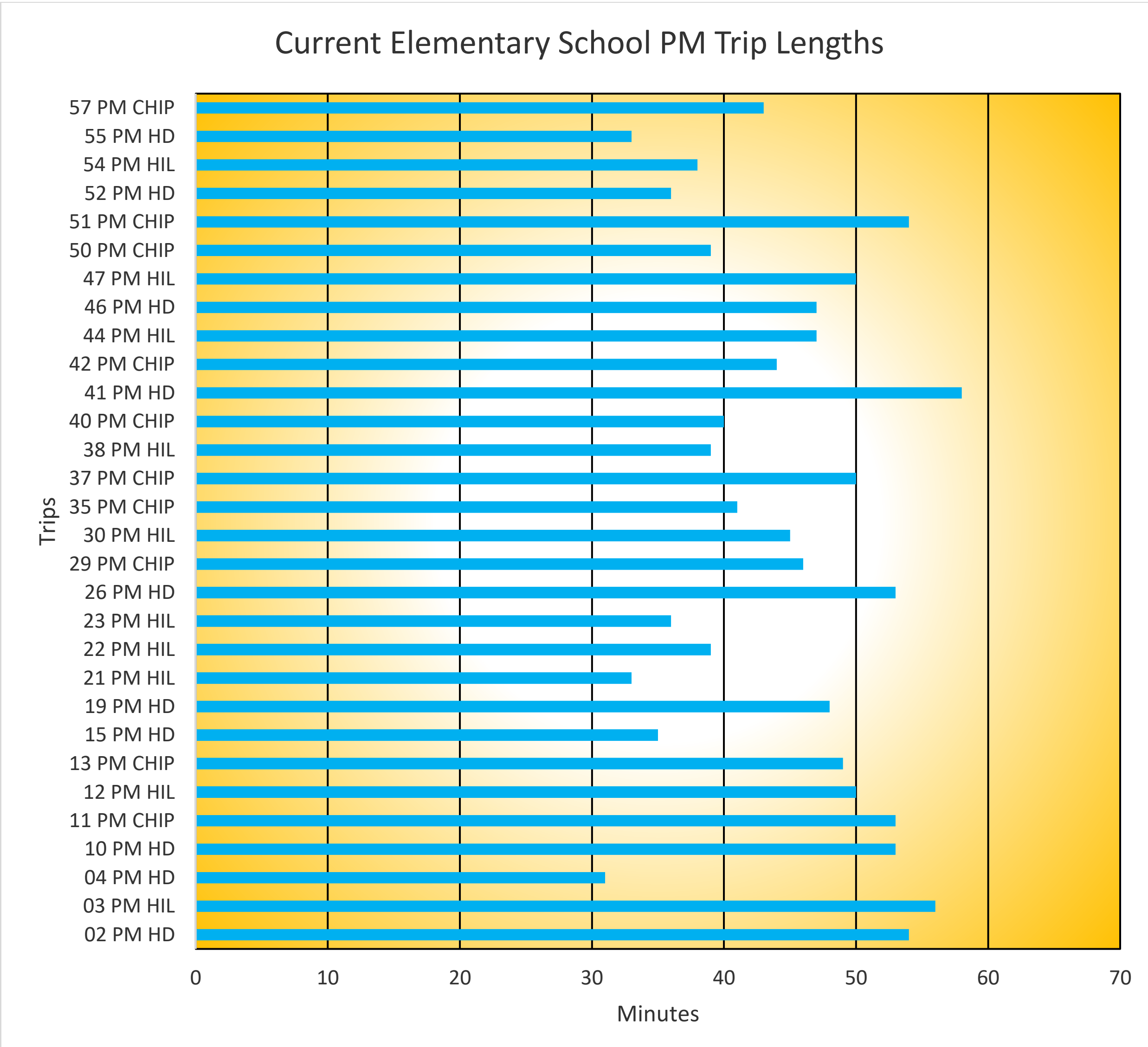
One High School bus run must travel to the far reaches of the district boundaries and then come back. This trip is so long it cannot take a Middle School run.

Similarly, the chart on the following page shows several time confilcts between the Middle School runs and the Elementary runs.



It should be noted that some of the PM Middle School bus runs currently arrive at the Elementary Schools a little late. Most days they are not so late as to be disruptive, but timing is very tight.

The last tier of service in the afternoon does not have conflicts because they will only be going to the bus garage when they are done with this final bus run.



Using the conflicts as shown in the preceding charts, the table below labeled Current Routing Structure, shows the number of buses needed based on the time conflicts as they are shown in the charts above.

Current Three-Tiered Bell Times

School Names	Arrive	AM Bell	PM Bell	Depart	Length of Day
High School	7:15	7:35	2:35	2:42	420 min
Middle School	8:00	8:10	3:05	3:15	415 min
Chippewa Elem School	8:55	9:10	3:40	3:50	400 min
Highland Elem School	8:55	9:10	3:40	3:50	400 min
Hilton Elem School	8:55	9:10	3:40	3:50	400 min

Current Three-Tiered Routing Structure

	Runs		Runs
Tier 1 Runs AM	23	Tier 1 Runs PM	26
Tier 1 SPED/CVCC Runs AM	3	Tier 1 SPED Runs PM	2
Tier 2 Runs operating	5	Private School Runs PM	5
		Preschool Runs PM	4
Total Buses Tier 1 AM	31	Total Buses Tier 1 PM	37
Tier 2 Runs AM	30	Tier 2 Runs PM	31
Tier 2 SPED Runs AM	1	Tier 1 Runs still operating	3
Private School Runs AM	4	Tier 2 SPED Runs PM	2
Preschool Runs AM	3	Private School Runs PM	3
Total Buses Tier 2 AM	38	Total Buses Tier 2 PM	39
Tier 3 Runs AM	29	Tier 3 Runs PM	30
Tier 3 SPED Runs AM	4	Tier 2 Runs still operating	2
Private School Runs AM	2	Tier 3 SPED Runs PM	4
		Private School Runs PM	3
Total Buses Tier 3	35	Total Buses Tier 3 PM	39
Expected Buses AM	38	Expected Buses PM	39

The table below provides a visualization for how individual bus runs are combined. It also shows how some bus runs may stretch beyond their expected time frame.

Bus	Capacity	AM 1	AM 2	AM 3	PM 1	PM 2	PM 3
2	71		MS	CHIP	HS	MS	HD
3	84	HS	PRIV	HIL	PRIV	MS	
4	71		MS	HIL	HS	MS	HD
5	18	CVCC	CVCC/PRE	SPED	PRE	SPED	SPED
6	16		PRE	SPED	PRE	SPED	SPED
7	71		MS	HIL	HS	MS	MS
9	84				HS	PRIV	PRIV
10	84	HS	MS	HD	HS	MS	HD
11	84	HS	MS	HD		MS	CHIP
12	84		MS	CHIP	HS	MS	HIL
13	84	HS	MS	HD	HS	MS	CHIP
14	20			SPED			SPED
15	84	HS	MS	CHIP	PRE	MS	HD
19	71		MS		PRIV	MS	HD
21	84	HS	MS	HD	CVCC	MS	HIL
22	84	HS	MS	CHIP	HS	MS	HIL
23	71	MS	MS		HS	MS	HIL
26	71	HS	MS	CHIP	HS	HS	HD
28	71	HS	MS	PRIV	PRE/HS	PRIV	PRIV
29	84	HS	MS	CHIP	HS	MS	CHIP
30	84	HS	MS	HD	HS	MS	HIL
35	71	HS	MS	CHIP	HS	MS	CHIP
37	71	MS	MS	CHIP	HS	MS	CHIP
38	84	HS	MS	HIL	HS	MS	HIL
40	84	HS	MS	HD	HS	MS	CHIP
41	84	HS	MS	HD	HS	MS	HD
42	71	HS	MS	CHIP	HS	MS	CHIP
43	16	SPED	PRE	SPED	SPED	SPED	SPED
44	71	HS	MS	HIL		MS	HIL
45	65		PRIV			PRIV	PRIV
46	65	HS	MS	CHIP	HS	MS	HD
47	65	HS	CVCC	HIL	HS	MS	HIL
48	84	HS	MS	CHIP	HS	MS	MS
50	84	HS	MS	HIL	HS	MS	CHIP
51	84	MS	MS	HD	HS	MS	CHIP
52	84	MS	MS	HD	HS	MS	HD
53	16	SPED	SPED	SPED	SPED	SPED	SPED
54	71	HS	PRIV	HIL	PRIV	MS	HIL
55	71	MS	MS	HIL		MS	HD
57	84	HS	MS		HS	MS	CHIP

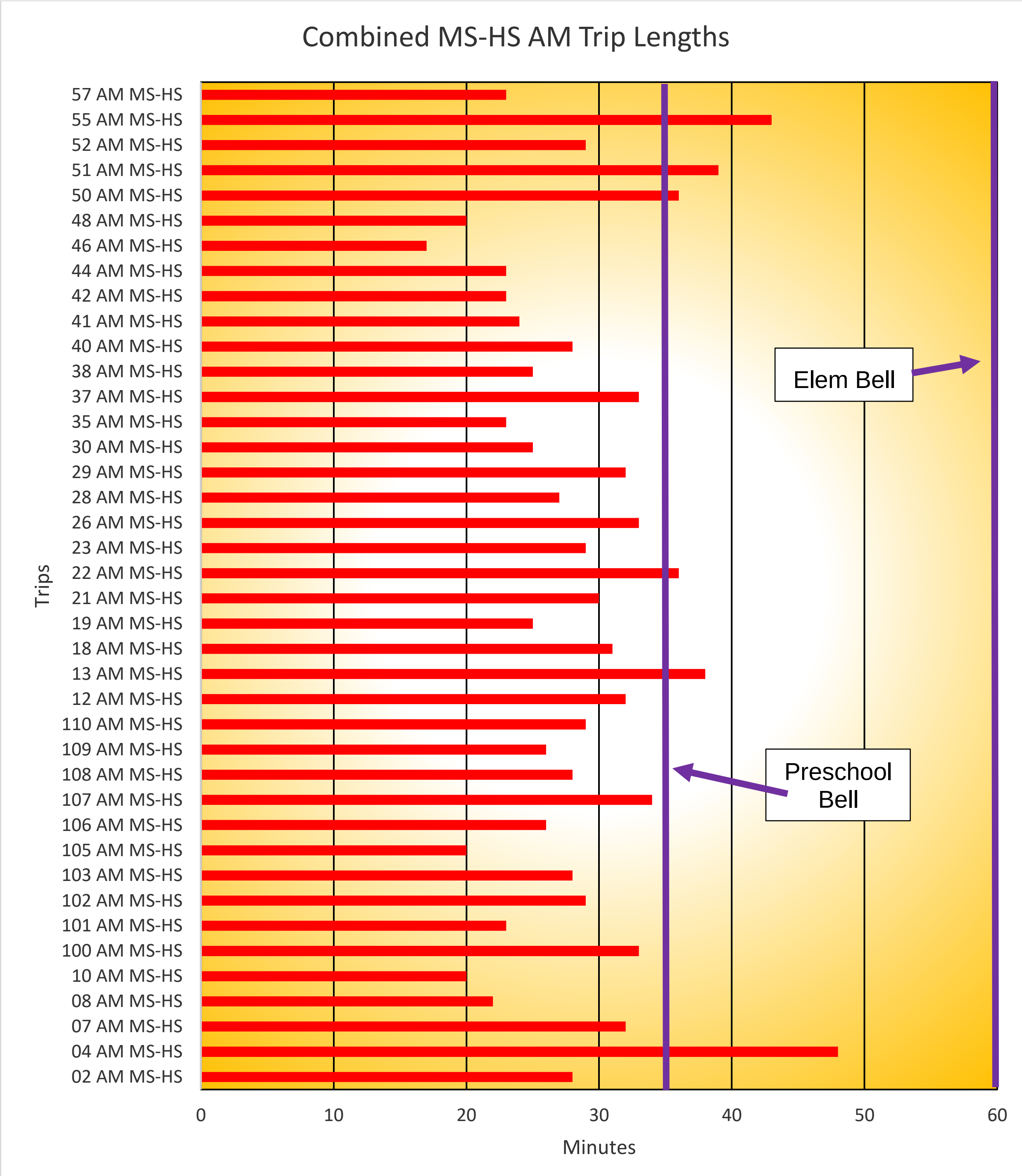
The above tables are provided as a baseline comparison to the planned routes for next year. BBHCSD is currently operating 39 buses for transporting students.

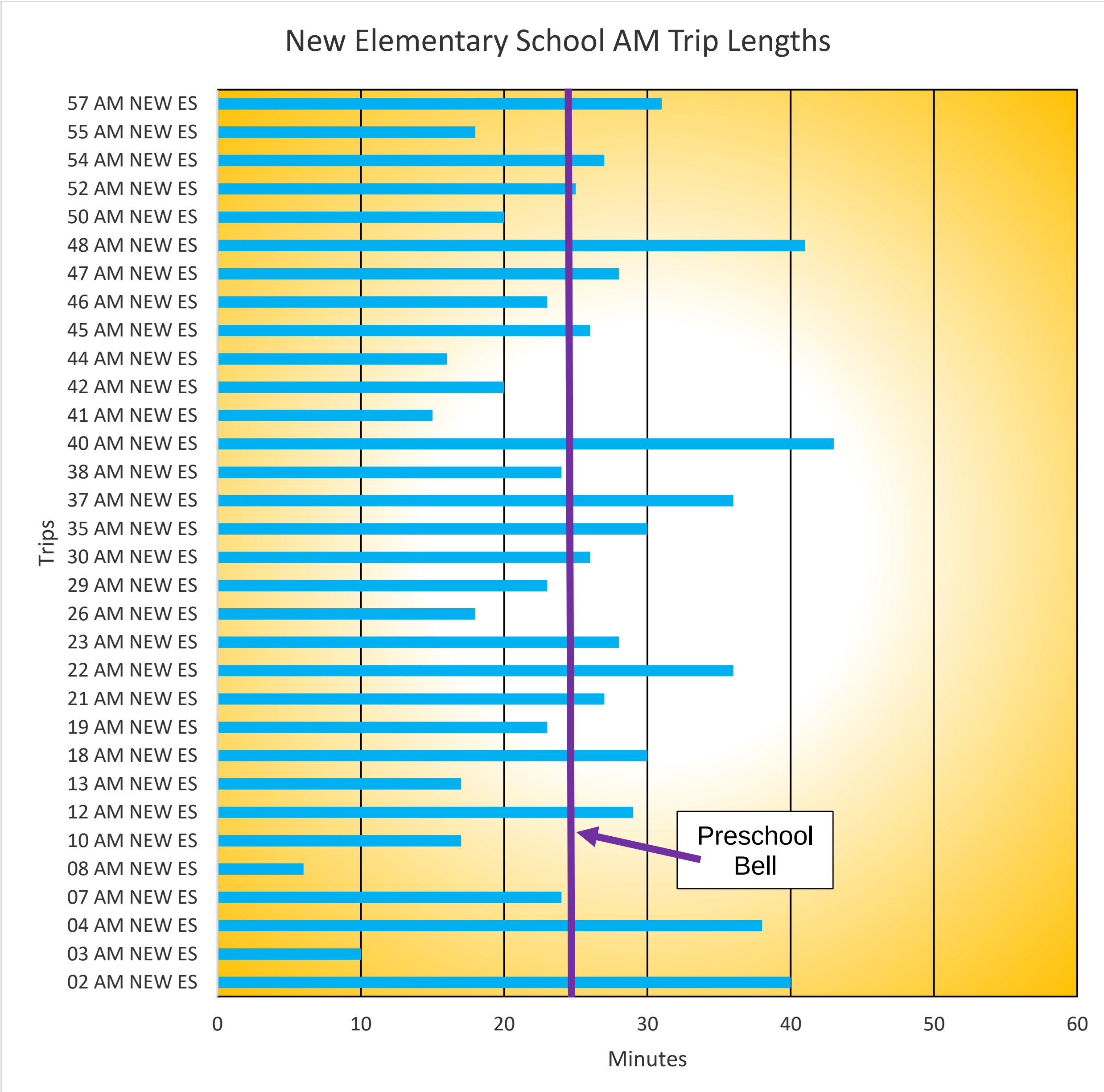
Proposed Bell Time Change

Brecksville-Broadview Heights City School District requested analysis be done on a particular bell time option for a proposed two-tiered routing structure with the Middle School and High School bus runs combined in a single tier and the Elementary bus runs adjusted to the new elementary school location. Below are the proposed bell times:

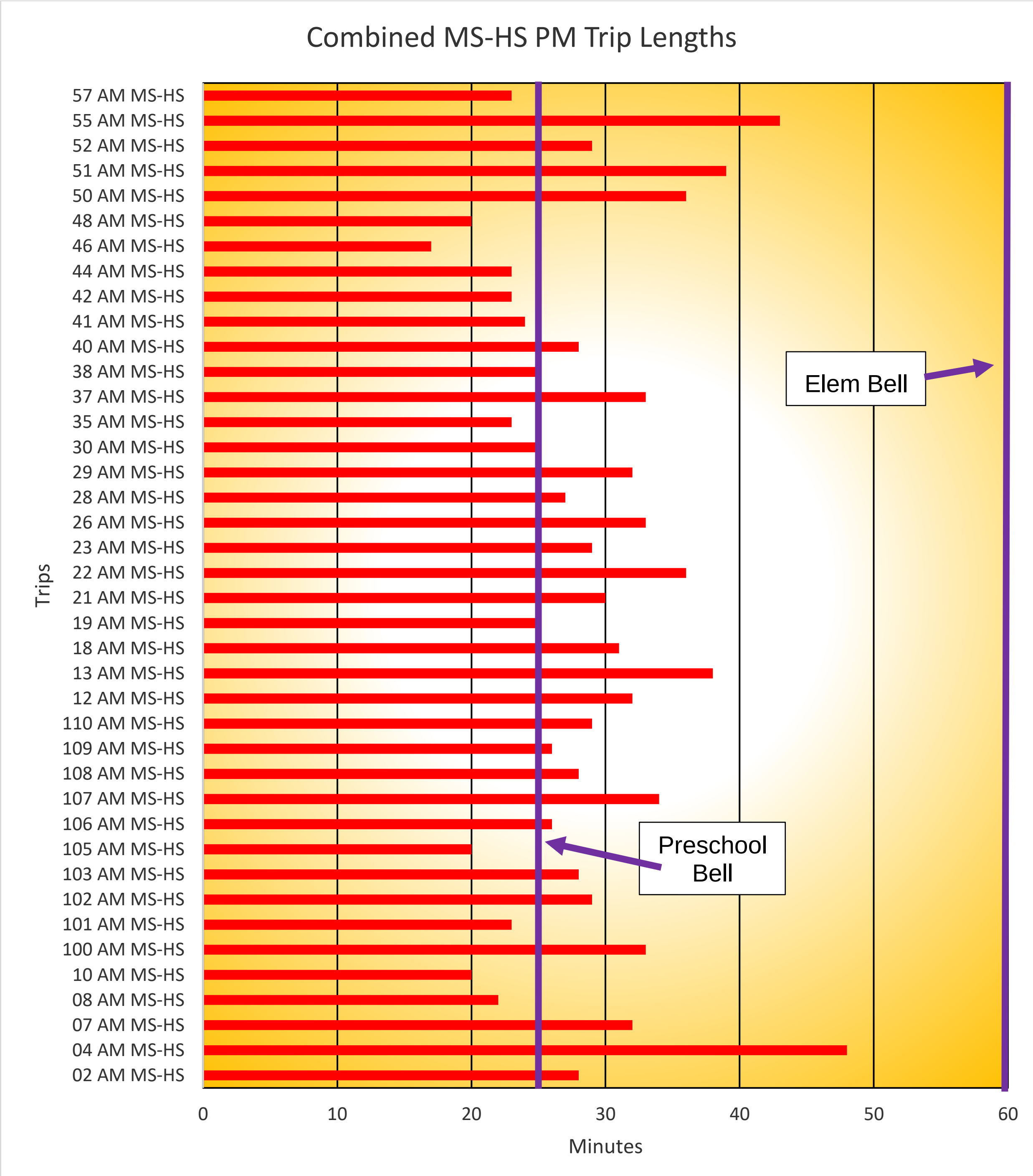
School Names	Arrive	AM Bell	PM Bell	Depart	Length of Day
Middle School & High School	7:25	7:45	2:45	3:00	420 min
New Elementary School	8:25	8:45	3:45	4:00	420 min
Preschool AM	8:00	8:20	10:50	11:00	150 min
Preschool PM	11:30	11:50	2:20	2:30	150 min

The following trip length charts show the length of the adjusted bus runs in time. You can compare these charts to the charts available on pages 15-20.

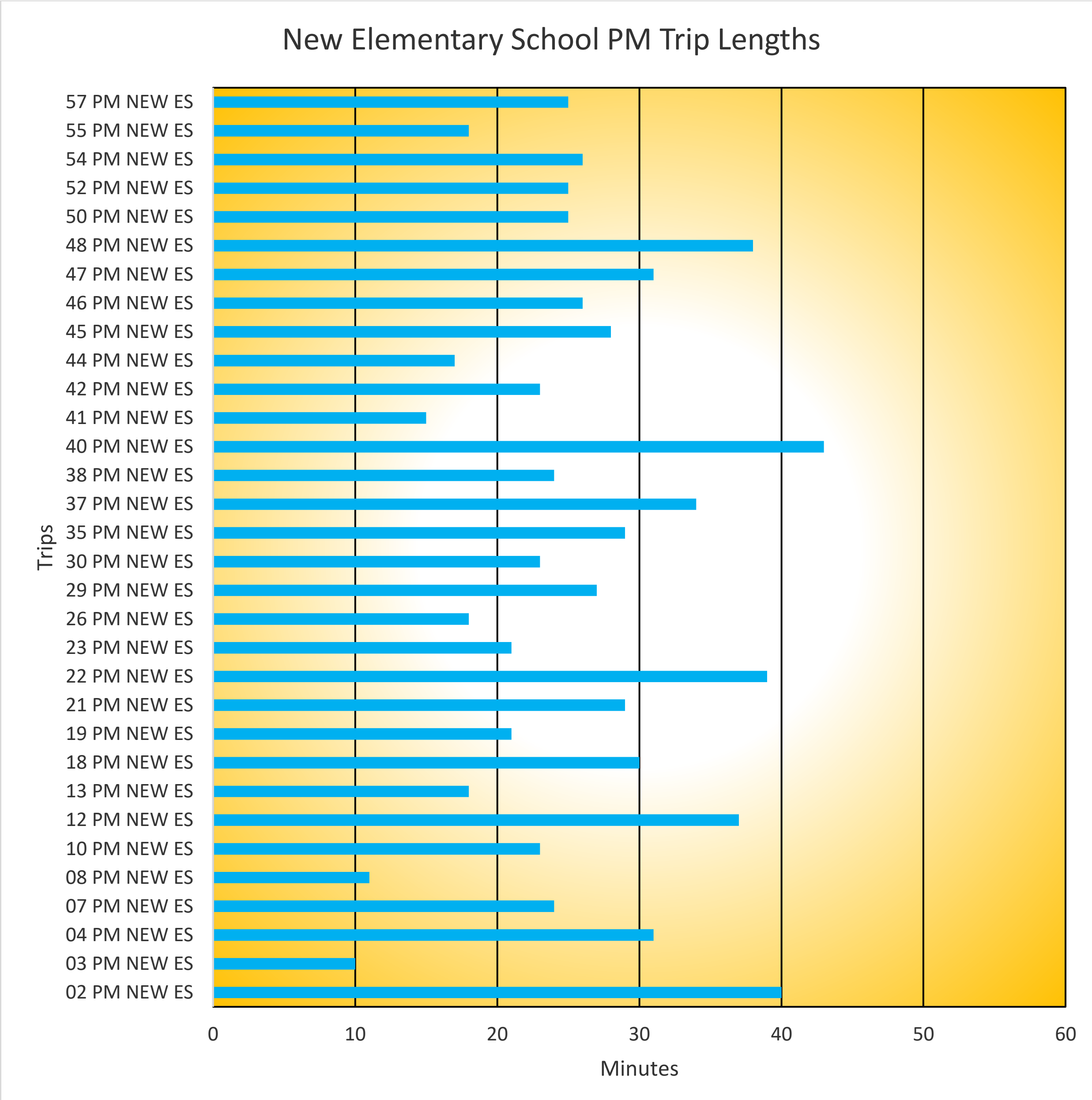

Combined MS-HS AM Trip Lengths
57 AM MS-HS
55 AM MS-HS
52 AM MS-HS
51 AM MS-HS
50 AM MS-HS
48 AM MS-HS
46 AM MS-HS
44 AM MS-HS
42 AM MS-HS
41 AM MS-HS
40 AM MS-HS
38 AM MS-HS
37 AM MS-HS
35 AM MS-HS
30 AM MS-HS
29 AM MS-HS
28 AM MS-HS
26 AM MS-HS
23 AM MS-HS
22 AM MS-HS
21 AM MS-HS
19 AM MS-HS
18 AM MS-HS
13 AM MS-HS
12 AM MS-HS
110 AM MS-HS
109 AM MS-HS
108 AM MS-HS
107 AM MS-HS
106 AM MS-HS
105 AM MS-HS
103 AM MS-HS
102 AM MS-HS
101 AM MS-HS
100 AM MS-HS
10 AM MS-HS
08 AM MS-HS
07 AM MS-HS
04 AM MS-HS
02 AM MS-HS
Trips
Elem Bell
Preschool Bell
0
10
20
30
40
50
60
Minutes



Note that the Preschool start time has conflicts with several of the new elementary school bus runs.



Note that the Preschool dismissal time has conflicts with the Middle School and High School bus runs.


New Elementary School PM Trip Lengths
57 PM NEW ES
55 PM NEW ES
54 PM NEW ES
52 PM NEW ES
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45 PM NEW ES
44 PM NEW ES
42 PM NEW ES
41 PM NEW ES
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13 PM NEW ES
12 PM NEW ES
10 PM NEW ES
08 PM NEW ES
07 PM NEW ES
04 PM NEW ES
03 PM NEW ES
02 PM NEW ES
Trips
0
10
20
30
40
50
60
Minutes

Proposed New Bell Times

School Names	Arrive	AM Bell	PM Bell	Depart	Length of Day
Middle School & High School	7:25	7:45	2:45	3:00	420 min
New Elementary School	8:25	8:45	3:45	4:00	420 min
Preschool AM	8:00	8:20	10:50	11:00	150 min
Preschool PM	11:30	11:50	2:20	2:30	150 min

Busing Resources Need for Proposed Two-Tiered Structure

	Runs		Runs
Tier 1 Runs AM	40	Tier 1 Runs PM	36
Tier 1 SPED/CVCC Runs AM	4	Tier 1 SPED/CVCC Runs PM	4
Private School Runs AM	3	Private School Runs PM	3
		Preschool Runs AM	3
Total Buses Tier 1 AM	47	Total Buses Tier 1 PM	46
Tier 2 Runs AM	32	Tier 2 Runs PM	33
Tier 2 SPED Runs AM	4	Tier 2 SPED Runs PM	4
Private School Runs AM	1	Private School Runs PM	2
Preschool Runs AM	3		
Total Buses Tier 2 AM	40	Total Buses Tier 2 PM	39
Expected Buses AM	47	Expected Buses PM	46

When including the additional vehicle needs for special needs routes, preschool routes and others, the planned routes for 2022-2023 will require 47 buses. This is 8 more buses than are currently being used.

The table below provides a visualization for how individual bus runs are combined.

Bus	Capacity	AM 1	AM 2		PM 1	PM 2
2	71	MS/HS	ELEM		MS/HS	ELEM
3	84	PRIV	ELEM		PRIV	ELEM
4	71	MS/HS	ELEM		MS/HS	ELEM
5	18		SPED		SPED	SPED
6	16	SPED	SPED		SPED	SPED
7	71	MS/HS	ELEM		MS/HS	ELEM
8	71	MS/HS	ELEM			ELEM
10	84	MS/HS	ELEM			ELEM
12	84	MS/HS	ELEM		MS/HS	ELEM
13	84	MS/HS	ELEM		MS/HS	ELEM
18	71	MS/HS	ELEM		PRE	ELEM
19	71	MS/HS	ELEM		MS/HS	ELEM
21	84	MS/HS	ELEM		MS/HS	ELEM
22	84	MS/HS	ELEM		MS/HS	ELEM
23	71	MS/HS	ELEM		MS/HS	ELEM
26	71	MS/HS	ELEM		MS/HS	ELEM
28	71	MS/HS	PRIV			PRIV
29	84	MS/HS	ELEM		MS/HS	ELEM
30	84	MS/HS	ELEM			ELEM
35	71	MS/HS	ELEM		MS/HS	ELEM
37	71	MS/HS	ELEM		MS/HS	ELEM
38	84	MS/HS	ELEM		MS/HS	ELEM
40	84	MS/HS	ELEM		MS/HS	ELEM
41	84	MS/HS	ELEM		MS/HS	ELEM
42	71	MS/HS	ELEM		MS/HS	ELEM
43	16	SPED	SPED		SPED	SPED
44	71	MS/HS	ELEM		MS/HS	ELEM
45	65	PRIV	ELEM		PRIV	ELEM
46	65	MS/HS	ELEM		MS/HS	ELEM
47	65	CVCC	ELEM		MS/HS	ELEM
48	84	MS/HS	ELEM		MS/HS	ELEM
50	84	MS/HS	ELEM		MS/HS	ELEM
51	84	MS/HS			MS/HS	PRIV
52	84	MS/HS	ELEM		MS/HS	ELEM
53	16	SPED	SPED		SPED	SPED
54	71	PRIV	ELEM		PRIV	ELEM
55	71	MS/HS	ELEM		MS/HS	ELEM
57	84	MS/HS	ELEM		MS/HS	ELEM
100	71	MS/HS			MS/HS	ELEM
101	71	MS/HS			MS/HS	
102	71	MS/HS			MS/HS	
103	71	MS/HS			MS/HS	
105	71	MS/HS			MS/HS	
106	71	MS/HS			MS/HS	
107	71	MS/HS			MS/HS	
108	71	MS/HS			MS/HS	
109	71	MS/HS			MS/HS	
110	71	MS/HS			MS/HS	

As you can see from the tables on the previous pages, the Middle School and High School bus runs are the main factor in determining the number of buses needed for the proposed bell times. Combining the Middle School and High School bus runs into a single tier required 40 buses in the morning. This, combined with Private School and Special Needs bus runs drives the number of required vehicles to 47 buses. This is reasonable given the unknown factor of ridership on the buses.

Once BBHCSD has collected ridership data, it is likely to expect further reductions may be possible. A reasonable number of reductions would be to reduce by five buses. The yellow box around the bottom five buses shown on the table on the preceding page represents the possible reductions related to likely ridership.

Driver and Bus Aide Pay Impact

The shift from three tiers to two tiers will have an impact on take home pay for both drivers and bus aides. Two tiers will mean a smaller number of hours for both types of employee. Below are tables that show the average amount of daily pay for drivers and bus aides.

Drivers average hourly rate: $24.74 Bus Aid average hourly rate: $19.98

Current Average Driver	Time	Current Average Bus Aide	Time
Pre Trip	30	Ave AM Tier 1 Runtime	38
Ave AM Tier 1 Runtime	38	Deadhead to Tier 2 stop	15
Deadhead to Tier 2 stop	15	Ave AM Tier 2 Runtime	30
Ave AM Tier 2 Runtime	30	Deadhead to Tier 3 stop	15
Deadhead to Tier 3 stop	15	Ave AM Tier 3 Runtime	31
Ave AM Tier 3 Runtime	31	Deadhead back to facility	15
Deadhead back to facility	15	Load time at HS	10
Sanitize bus	15	Ave PM Tier 1 Runtime	31
Load time at HS	10	Deadhead to Tier 2 stop	15
Ave PM Tier 1 Runtime	31	Load time at MS	10
Deadhead to Tier 2 stop	15	Ave PM Tier 2 Runtime	32
Load time at MS	10	Deadhead to Tier 3 stop	15
Ave PM Tier 2 Runtime	32	Load time at Elem	10
Deadhead to Tier 3 stop	15	Ave PM Tier 3 Runtime	45
Load time at Elem	10	Deadhead back to facility	15
Ave PM Tier 3 Runtime	45	**Total minutes**	**327**
Deadhead back to facility	15	**Total hours**	**5.45**
Post Trip	15	**Ave Daily pay**	**$108.89**
Attendance Tracking	10		
Total minutes	**397**		
Total hours	**6.62**		
Ave Daily pay	**$163.70**		

The table on the following page shows the possible average amount of daily pay for drivers and bus aides in a two-tiered schedule.

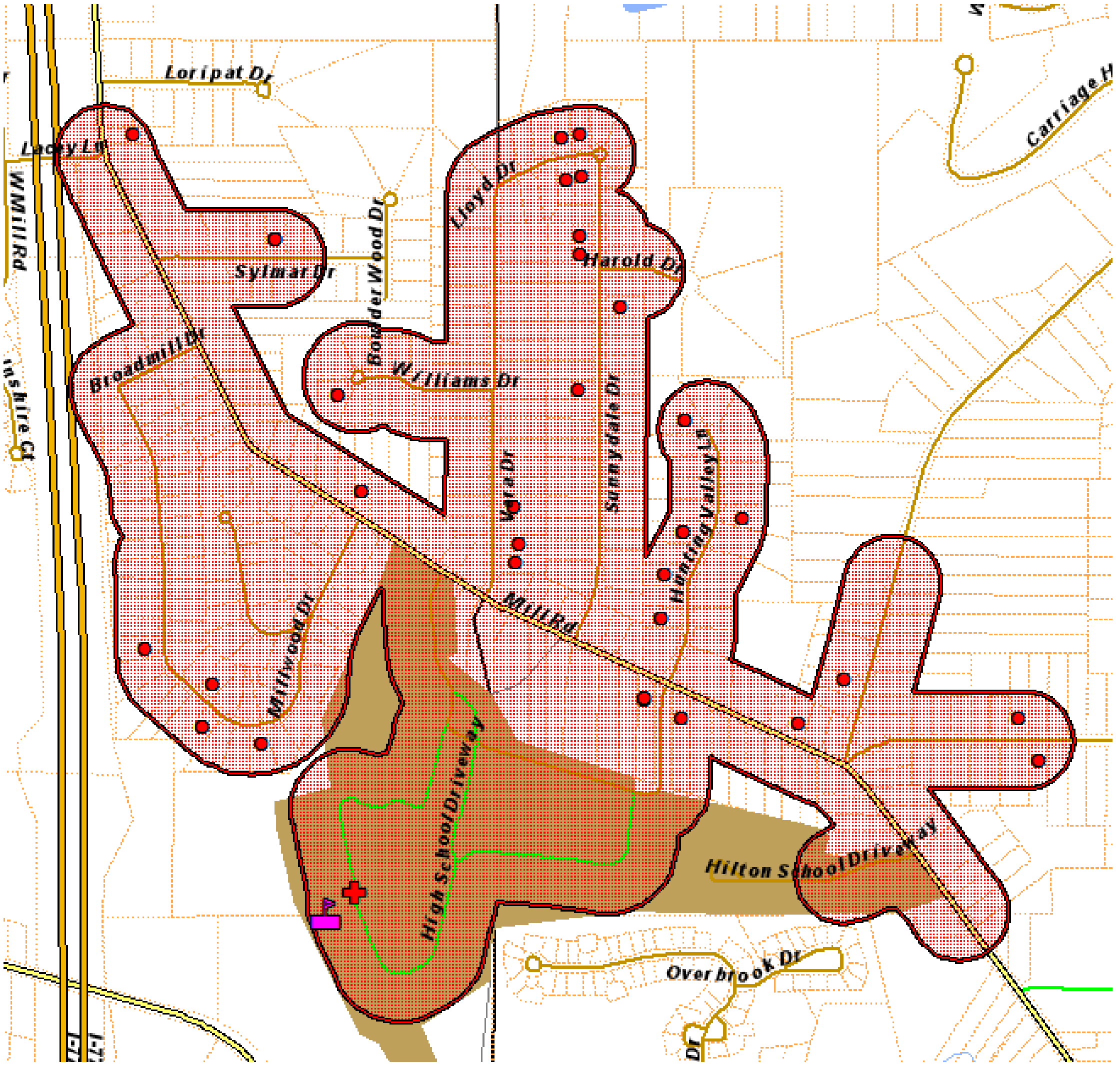
Possible Average Driver	Time	Possible Average Bus Aide	Time
Pre Trip	30	Ave AM Tier 1 Runtime	29
Ave AM Tier 1 Runtime	29	Deadhead to Tier 2 stop	15
Deadhead to Tier 2 stop	15	Ave AM Tier 2 Runtime	25
Ave AM Tier 2 Runtime	25	Deadhead back to facility	15
Deadhead back to facility	15	Load Time at MS/HS	15
Sanitize bus	15	Ave PM Tier 1 Runtime	30
Load Time at MS/HS	15	Deadhead to Elem	15
Ave PM Tier 1 Runtime	30	Load Time at Elem	15
Deadhead to Elem	15	Ave PM Tier 2 Runtime	26
Load Time at Elem	15	Deadhead back to facility	15
Ave PM Tier 2 Runtime	26	**Total minutes**	**200**
Deadhead back to facility	15	**Total hours**	**3.33**
Post Trip	15	**Ave Daily pay**	**$66.60**
Attendance Tracking	10		
Total minutes	**270**		
Total hours	**4.50**		
Ave Daily pay	**$111.33**		

As you can see, the switch from a three-tiered routing schedule to a two-tiered schedule will substantially decrease transportation employee pay. **This will negatively impact the district's ability to retain current employees as well as recruit new employees.**

Walk Zone Analysis

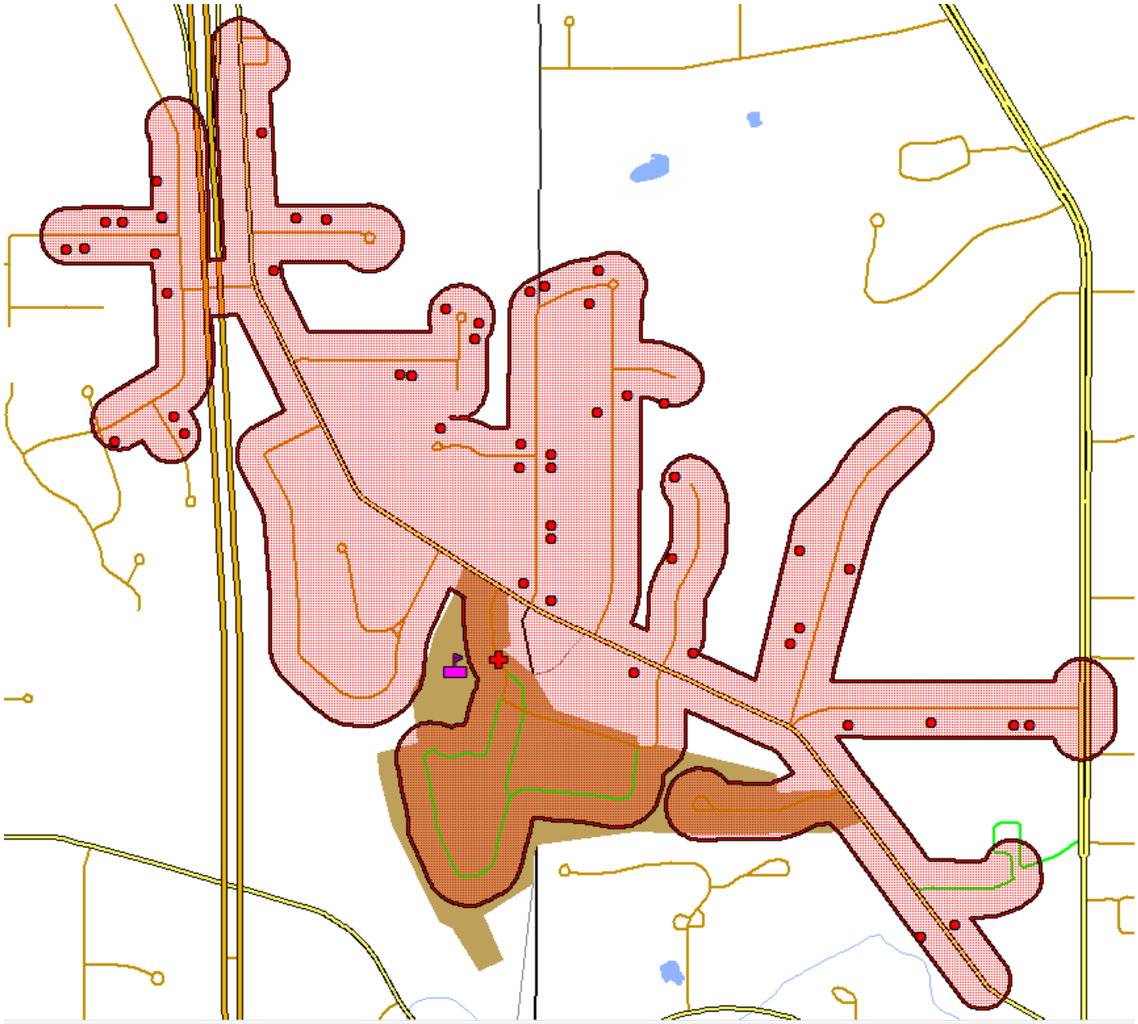
<u>High School Walk Zone</u>

The image below depicts a walk zone for the High School if it were to be drawn to include all areas exactly one mile from the school. Using the current enrollment this would impact a total of 40 students, six of whom are private school students.



Middle School Walk Zone

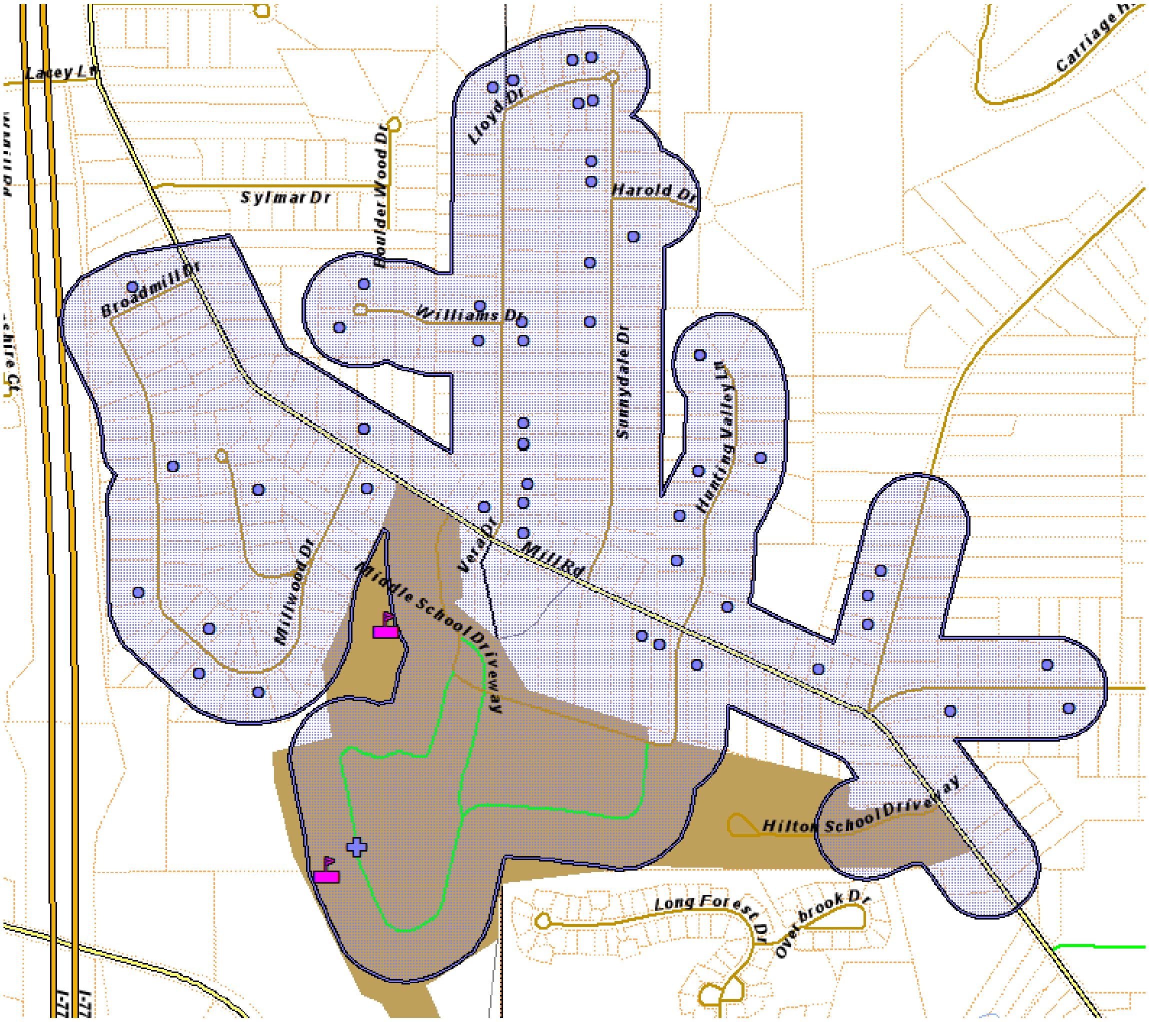
The image below depicts a walk zone for Middle School if it were to be drawn to include all areas exactly one mile from the school. Using the current enrollment this would impact a total of 65 students, four of whom are private school students.



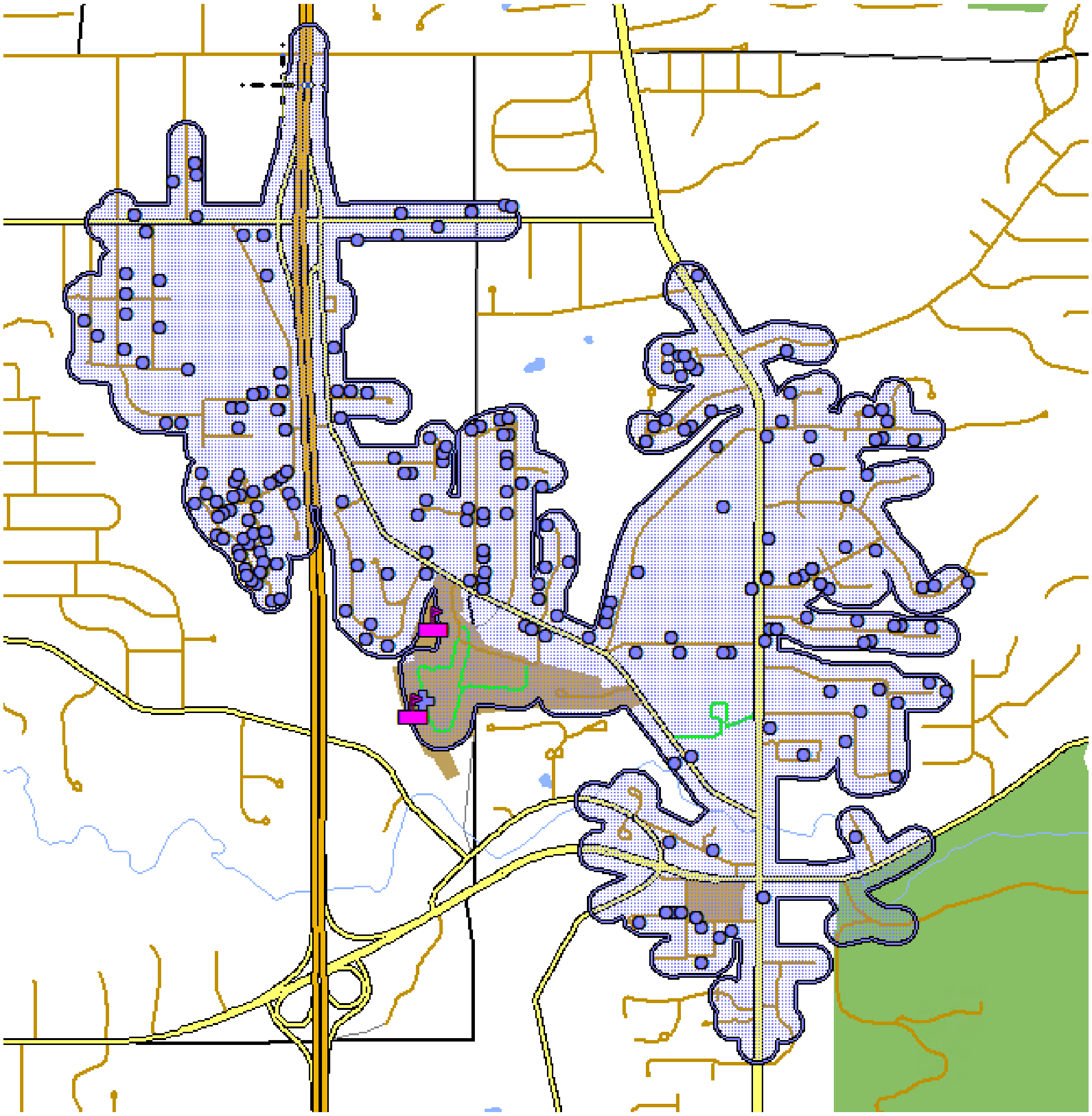
It should be noted that some of the areas shown as part of the 1-mile walk zone for the Middle School would involve students walking over a bridge that crosses the interstate as well as walking along sections of street that do not have sidewalks. This would not be considered best practice for Middle School students.

<u>Combined Middle School & High School Walk Zone</u>

A compromise solution might be to match the Middle School Walk Zone to that of the High School. The image below shows as example of this. Currently such a boundary would impact 36 Middle School Students, 34 High School Students and 3 Private School students. The impact of this change would be relatively small, but it may allow the district to reduce secondary routes by one bus run.



For comparison, below is an image of what a two mile walk zone around the Middle School and High School would look like.

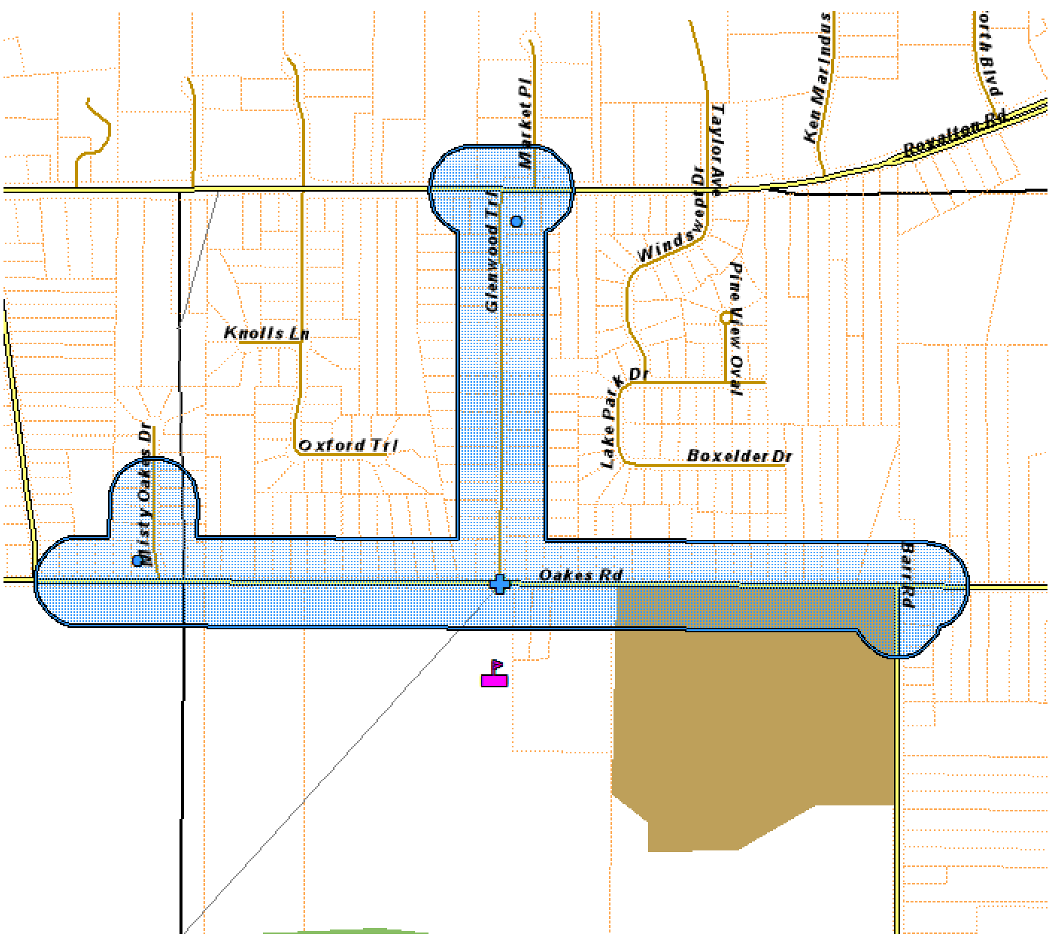


This option would include a considerably larger number of students. There are currently 130 Middle School Students, 135 High School Students, and 13 private school students living in this region for a total of 278 students. A walk zone of this area has the potential to reduce the total number of buses needed to serve the proposed two-tiered Middle School and High School tier by up to six buses.

It should be noted that this two-mile area would force many students to walk over a bridge that crosses the interstate as well as walk along some areas with heavy amounts of traffic. These areas would be considered hazardous, and it would be recommended to not include them in a walk zone around the two schools.

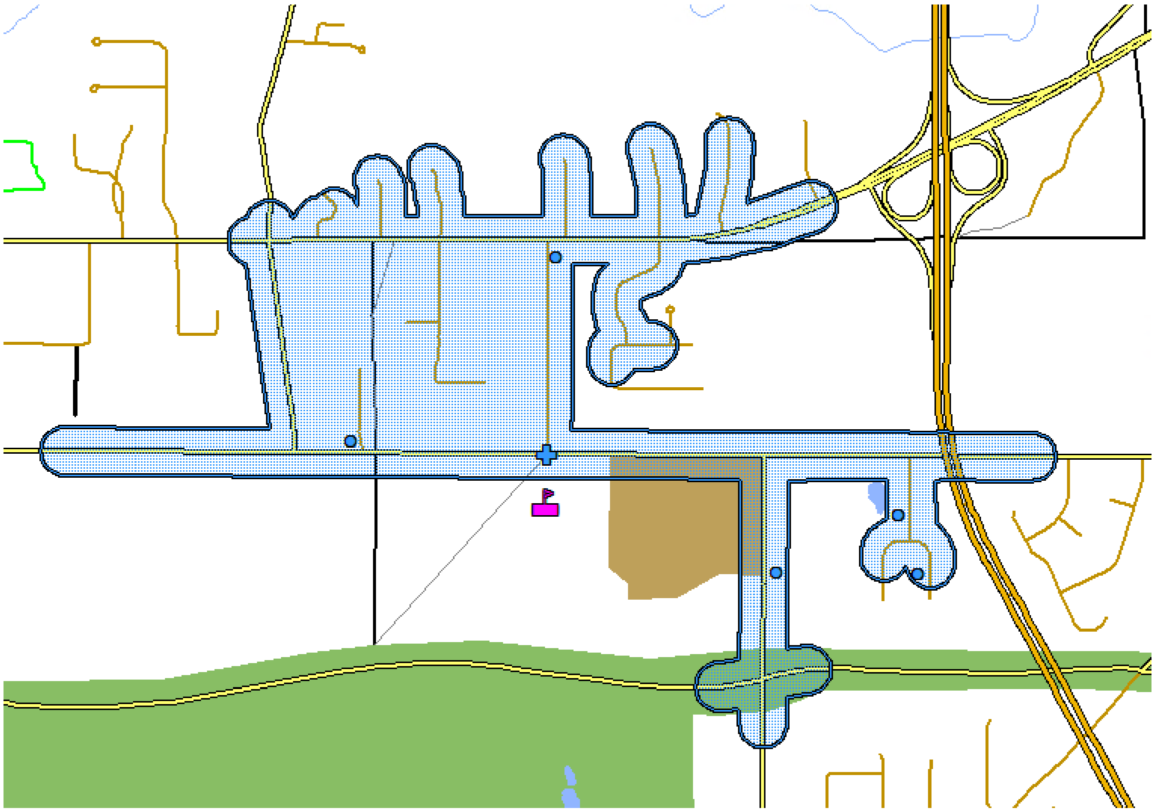
New Elementary School Walk Zone

The current elementary schools do not have a walk zone. BBHCSD has considered a half mile walk zone be instituted for Elementary students in the future. Unfortunately, there are very few houses in close proximity of the new elementary school location. Only two students currently live within a half mile of the new school. The image below depicts the half mile area around the school.



At this time a half mile walk zone for the new elementary school would serve no purpose.

For comparison the image below shows what a one-mile walk zone would look like for the new elementary school



A one-mile zone would currently impact seven students. However, five of those students are on cross streets that would require the students to walk some distance along Oakes Rd with little protection from traffic.

Executive Summary

Transfinder's analysis of BBHCSD's current bus routes indicated that the district's routes are relatively efficient.

Transfinder analyzed the following proposed two-tier bell time option along with a revised set of routes.

School Names	Arrive	AM Bell	PM Bell	Depart	Length of Day
Middle School & High School	7:25	7:45	2:45	3:00	420 min
New Elementary School	8:25	8:45	3:45	4:00	420 min
Preschool AM	8:00	8:20	10:50	11:00	150 min
Preschool PM	11:30	11:50	2:20	2:30	150 min

Transfinder found the proposed two-tiered bus routes are also relatively efficient with the planned routes having even higher assigned ridership than the current three-tiered routes. This is mostly possible due to the increase amount of time between tiers which allows more time to fill the bus. The number of buses needed to support the proposed two-tier times would increase buses from 39 to 47 for planning purposes. It is possible that reductions could be found in subsequent years following the change in tiers after ridership numbers on the number of likely riders have been collected for the new configuration.

Changing from three tiers to two tiers will reduce the total number of hours that drivers and bus aides will have available to work. This will decrease the total take-home pay that those employees receive. Decreased hours and pay may make it hard for BBHCSD to retain current employees and to recruit new employees in an already difficult market to attract drivers.

A review of the proposed standardization of 1 mile around the High School and Middle School and one-half mile around the New Elementary School showed little gain to be found by making the proposed change. A compromise which would match the Middle School walk zone and the

High School walk zone would affect 73 students and could reduce the vehicle needs by one bus. An example of a two-mile walk zone around the Middle School and High School was provided, but not recommended for safety reasons. A half mile walk zone around the new elementary school would have practically no impact at all as there are only two elementary students living in that area. A one-mile walk zone around the new elementary school was provided, but not recommended for safety reasons.

The following pages outline some of the Professional Services offered by Transfinder that may be helpful to your district.

Other Transfinder Professional Services available:

We provide implementation services for the full suite of our solutions, including our fleet maintenance, field trip, and AVL software for seamless GPS integration with our routing system.

1. **New Route Creation Based on Changing Attendance Patterns**
   Demographic shifts and student matriculation changes often result in the opening of a new school or the consolidation or closing of schools. This dramatically affects a district's routing and scheduling landscape. We review your new data to assess the impact on your transportation needs; create new routes and bus runs; and designate appropriate bus stops to accommodate change; and reduce mileage and ride times.

2. **Continuous Routing Services**
   Our Professional Services staff is available throughout the country to provide routing services, which include (1) yearly route updates to improve efficiencies; (2) continuous route updates as needed; and (3) a subscription service that allows you to call upon our expertise as needed. We offer cost-effective solutions for unexpected changes in budget or personnel that ensure consistency in your operation.

MANAGEMENT TRAINING

Several of our Professional Services staff have managed transportation departments and can provide management training for directors throughout the country. We assess your day-to-day operational challenges and provide customized management training to ensure your success.

TRANSPORTATION CONSOLIDATION STUDIES

Economic challenges are causing school districts to evaluate consolidating services, including transportation. Our Professional Services staff has worked with school districts in several states on opportunities for consolidation that ensure local control, while leveraging centralized routing and scheduling expertise. Our studies enable participating school districts to achieve economies of scale through careful analysis, collaborations, and agreements.